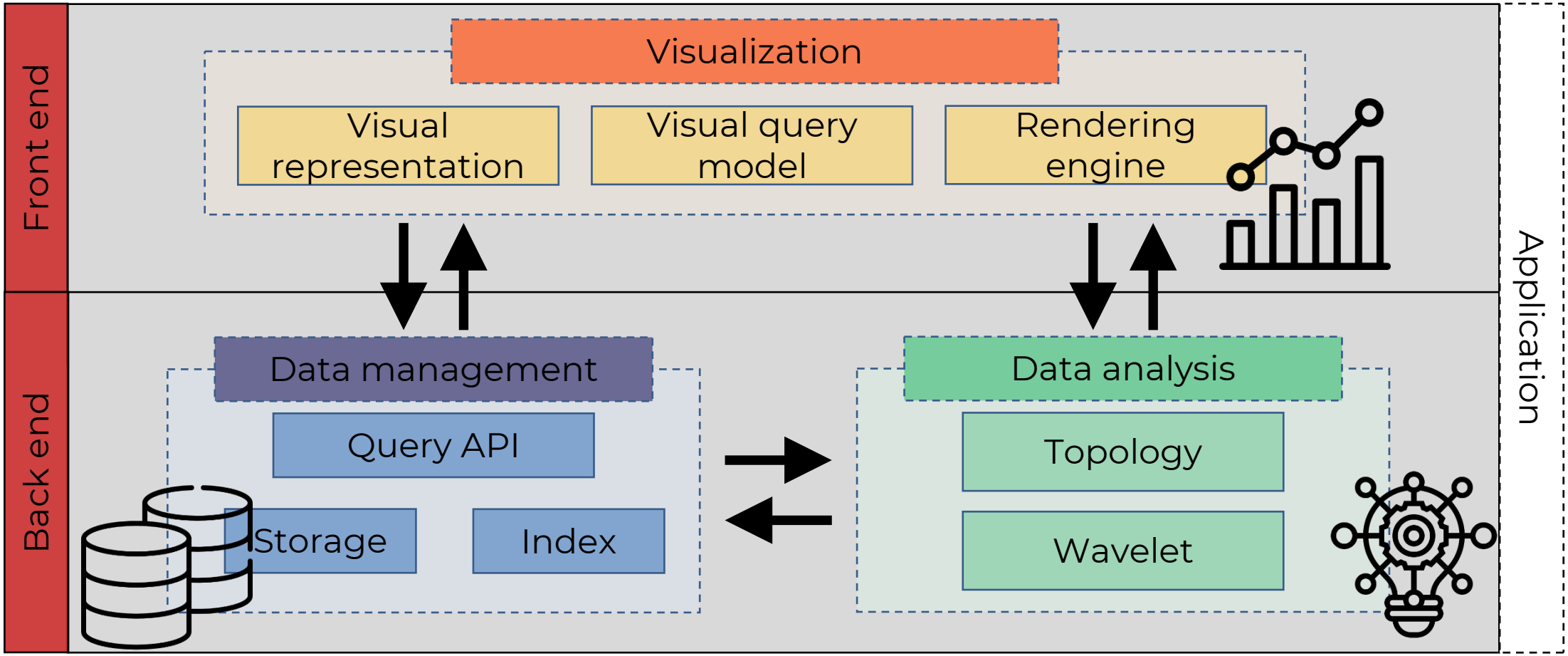
Visual analytics system building

CS524: Big Data Visualization & Analytics

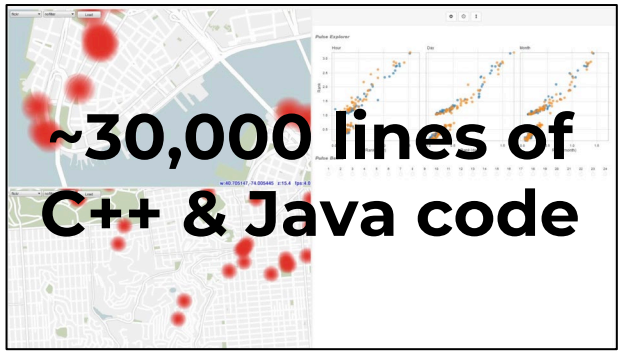
Fabio Miranda

Visual analytics system

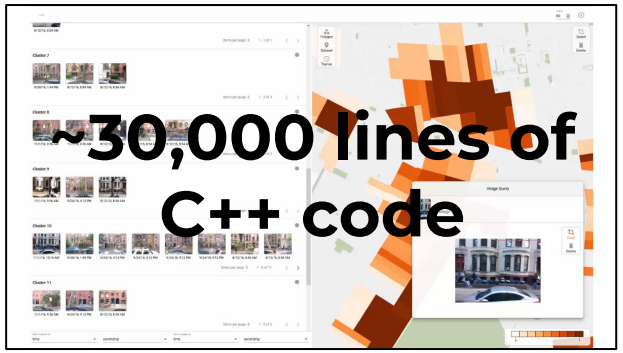


Building VA systems:
Understand → Ideate → Make → Evaluate

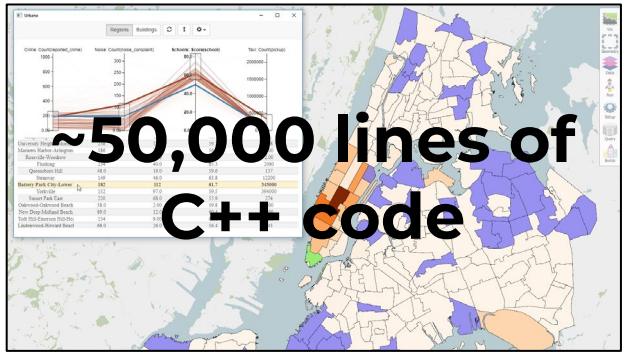
- Real-world problems are becoming more complex, and so are the data and VA systems used to tackle them.
- How are these systems built in practice?



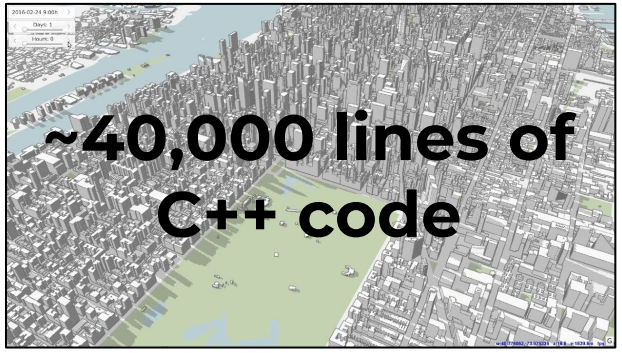
Urban Pulse:
Large-scale data mining of social media data



Urban Mosaic:
Interactive exploration of large imagery data



Urbane:
Interactive exploration of large data



Shadow Profiler:
City-scale assessment of sunlight access

Building VA systems:
Understand → Ideate → Make → Evaluate

- Real-world problems are becoming more complex, and so are the data and VA systems used to tackle them.
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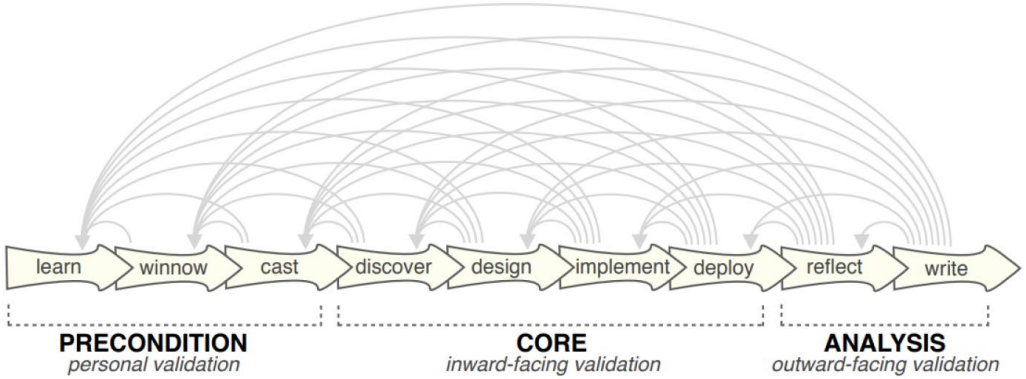
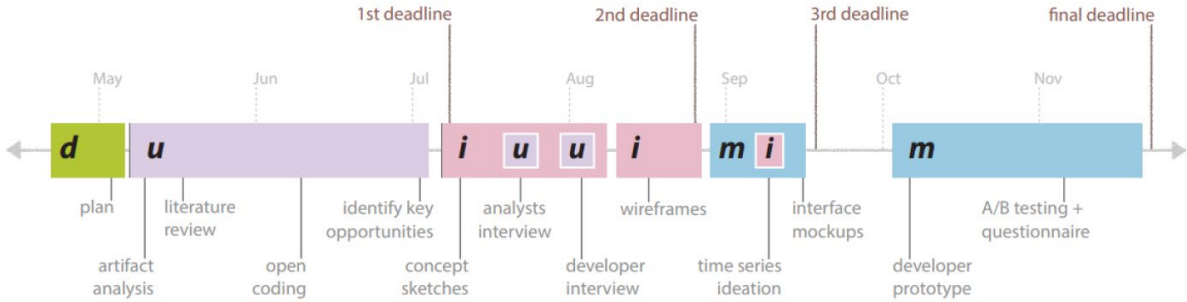
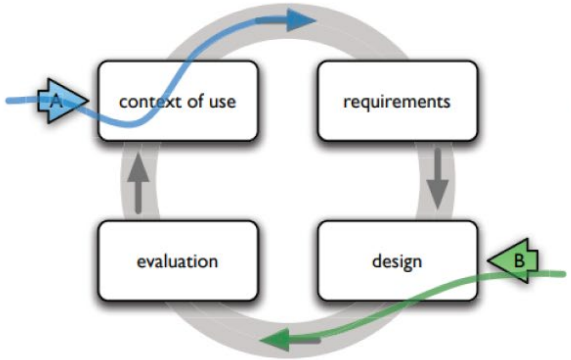
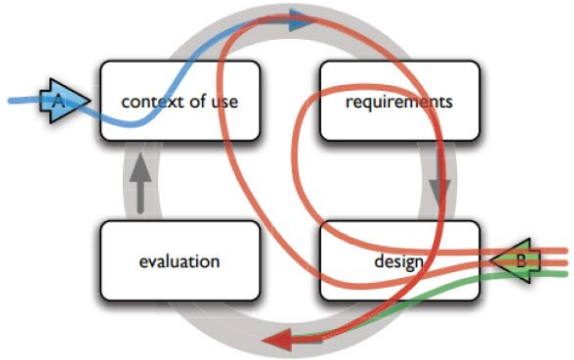
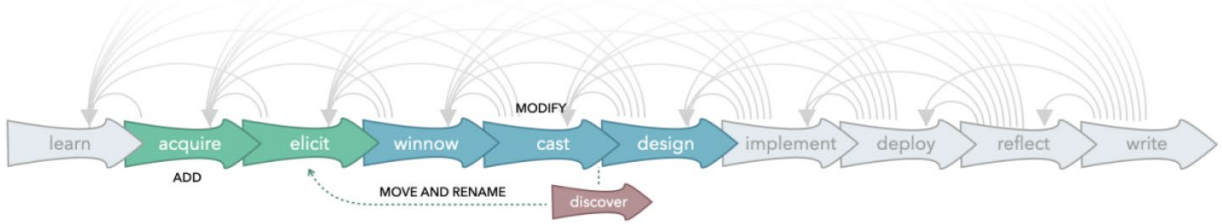






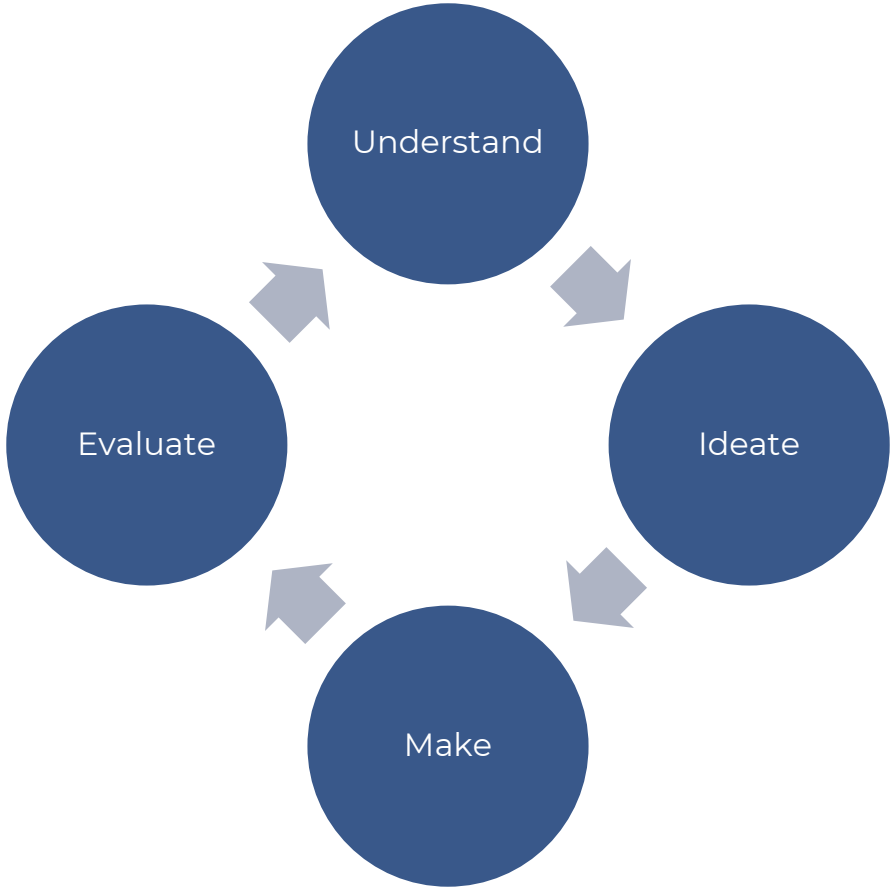




Building VA systems: Understand → Ideate → Make → Evaluate

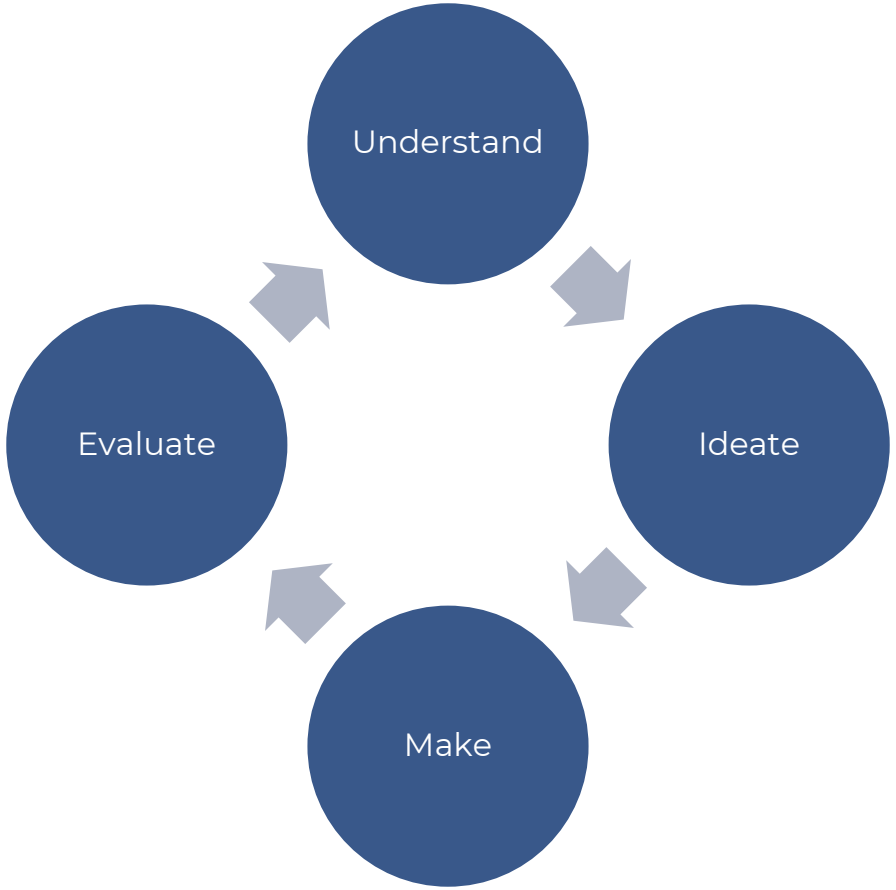
- Real-world problems are becoming more complex, and so are the data and VA systems used to tackle them.
- How are these systems built in practice?
  - Lloyd and Dykes, 2011:
    - Context of use → Requirements → Early prototypes → Later prototypes → Full application → Evaluation
  - Sedlmair et al., 2012:
    - Learn → Winnow → Cast → Discover → Design → Implement & deploy → Reflect & write
  - McKenna et al., 2014:
    - Understand → Ideate → Make → Deploy
  - Hall et al., 2019:
    - Data analysis → Study methods → Prototyping → Learning → Communicating
  - Oppermann and Munzner, 2020:
    - Learn → Acquire data → Elicit tasks → Winnow → Cast → Design → Implement & deploy → Reflect & write
  - Software engineering:
    - Requirements gathering → Design → Development → Testing → Deployment & maintenance

Building VA systems:
Understand → Ideate → Make → Evaluate



Important: design study methodologies can usually be mapped to this four-stage process.

Building VA systems:
Understand → Ideate → Make → Evaluate



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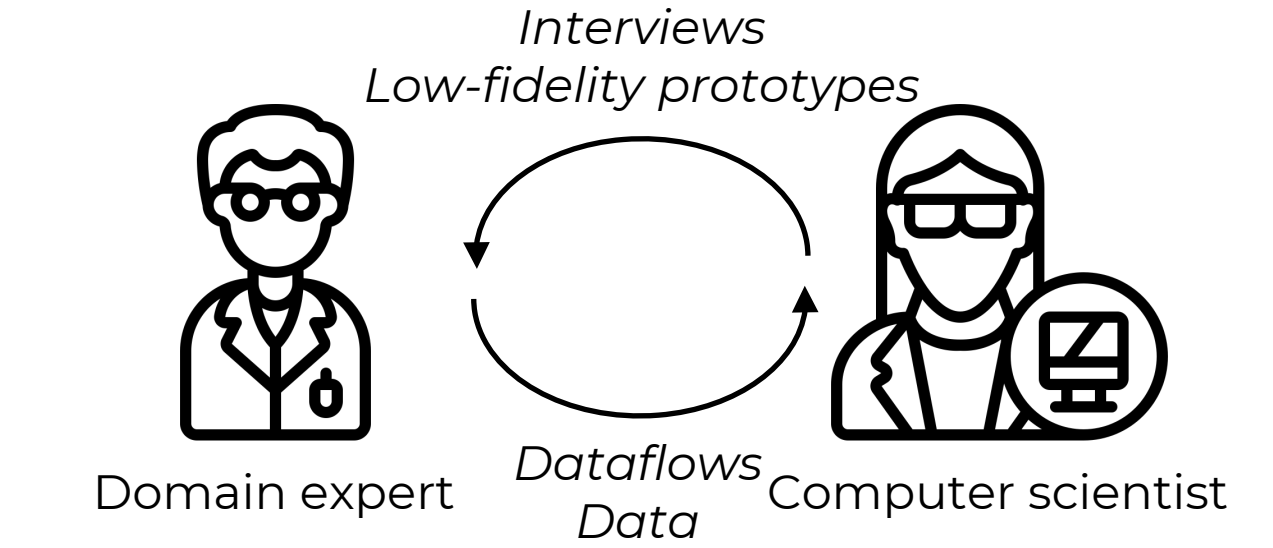
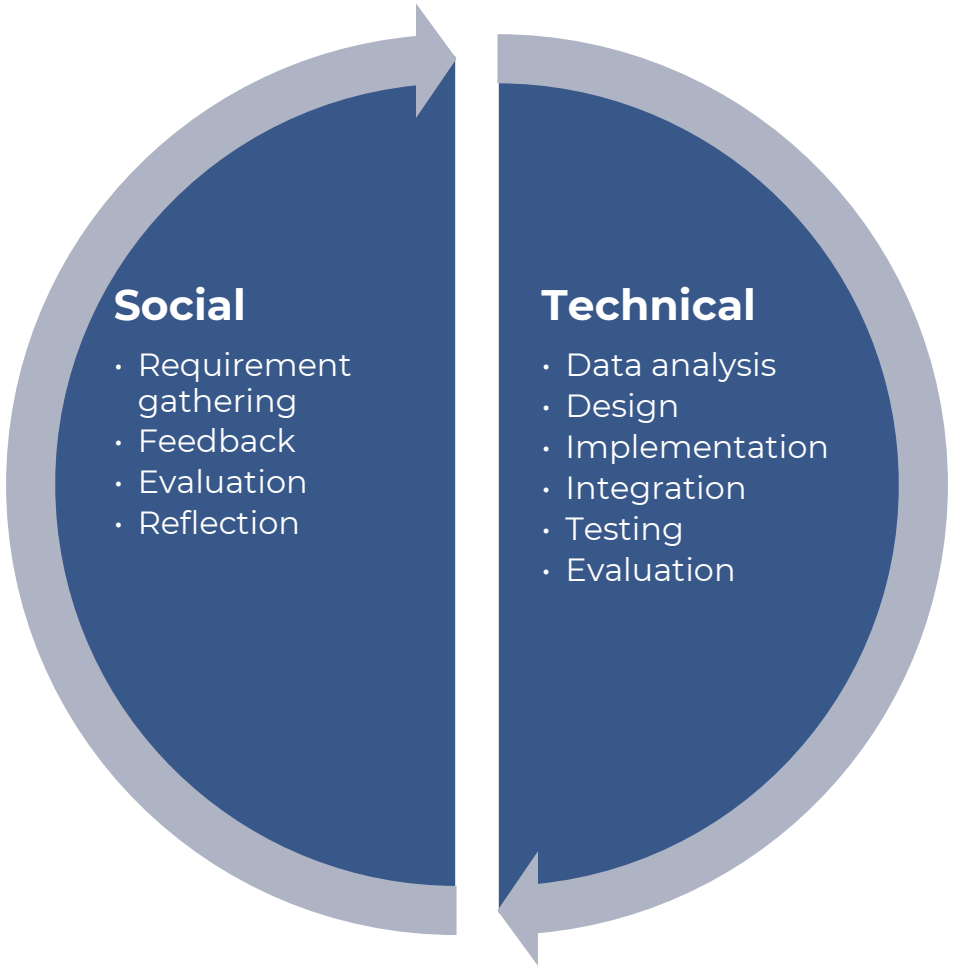


Domain knowledge

Computer science knowledge

Building VA systems:
Understand → Ideate → Make → Evaluate



Building VA systems:
Understand → Ideate → Make → Evaluate

Building VA systems:
Understand → Ideate → Make → Evaluate

- Building VA systems has been limited by the lack of “structured” guidance for how systems are composed and implemented.
- **AI is shifting the bottleneck:**

From *“Can we build it?”*

To *“Can we explore & justify choices and iterate faster without losing rigor?”*

New approaches for VA system building

Visualizationary: Automating Design Feedback for Visualization Designers Using Large Language Models

Sungbok Shin, Sanghyun Hong, and Niklas Elmqvist, *Fellow, IEEE*

***Abstract*—Interactive visualization editors empower users to author visualizations without writing code, but do not provide guidance on the art and craft of effective visual communication. In this article, we explore the potential of using an off-the-shelf large language models (LLMs) to provide actionable and customized feedback to visualization designers. Our implementation, VISUALIZATIONARY, demonstrates how ChatGPT can be used for this purpose through two key components: a preamble of visualization design guidelines and a suite of perceptual filters that extract salient metrics from a visualization image. We present findings from a longitudinal user study involving 13 visualization designers—6 novices, 4 intermediates, and 3 experts—who authored a new visualization from scratch over several days. Our results indicate that providing guidance in natural language via an LLM can aid even seasoned designers in refining their visualizations.**

***Index Terms*—Visualization design, design critique, feedback, human-centered AI, large language models.**

I. INTRODUCTION

TO DEMOCRATIZE the transformative power of data in today's information society, it is not sufficient that people get access to visualizations created by others; they must also be empowered to author their *own* visualizations. In response, recent years have seen an influx of interactive tools that enable users to create visualizations without writing code, such as iVisDesigner [51], Data Illustrator [38], and Lyra [56]. However, no matter how advanced these tools become, a fundamental barrier remains: authoring effective visualizations is a complex task requiring visual design skills, an understanding of aesthetics, and experience—expertise that not all would-be designers possess.

Fortunately, visualization designers have nothing to lose but their chains. We introduce VISUALIZATIONARY, a system to seize the means of visualization design by using off-the-shelf large-language models (LLMs). Our system enhances the design of communicative visualizations by providing perceptual feedback. While it may not fully replace feedback from peers or colleagues, it serves as a valuable alternative when it is impossible to gain human feedback and can benefit designers of all expertise levels, from novices to seasoned ones.

Although competing approaches for improving visualization designs, such as heuristics (like small, smart defaults in Lyra) and Grammar of Graphics (GoG)-style specifications have their own advantages, they also carry limitations due to their deterministic nature. Heuristic-driven methods lack human intervention, which can constrain creativity and limit the potential for novel ideas. GoG-style specifications, despite offering precise chart descriptions, lead to producing standardized solutions, reducing the diversity of possible visualization outcomes.

In contrast, Visualizationary automates perceptual feedback using a suite of automated techniques that simulate how a viewer would perceive a visualization. By providing actionable insights without the need to standardize changes, Visualizationary preserves the designer's creativity while enhancing effectiveness.

The basic idea behind Visualizationary is to investigate how an LLM can be used to aid the iterative design process of a visualization artifact in a workflow that we call *analyze-clarify-guide-track* (ACGT):

1) **Analyze** the state of a visualization using automated perceptual filters, yielding a corresponding analysis report;
2) **Clarify** the results from the filters in the report such that it is understandable to even novice visualization designers;
3) **Guide** the designer in making changes that address identified concerns in their visualization; and
4) **Track** the trajectory of the visualization artifact over time during its iterative design process.

We present a web-based implementation of Visualizationary that leverages a server-side LLM, demonstrating how a "vanilla" LLM can be used for this highly complex visualization design task. Visualizationary operates as follows: We first translate chart images into text using a vision-language model (VLM) [36] and then generate automatic design guidance using an LLM as a template and text processor. Users can also upload their existing

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by University of Maryland College Park (UMCP), College Park, MD Institutional Revew Board under Application No. 1185917-8 and Exemption category #45CFR46.104(d)(2)(i-ii).

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All our supplemental materials are available at https://osf.io/v7hu8.

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From Requirement to Solution: Unveiling Problem-Driven Design Patterns in Visual Analytics

Yuchen Wu, Shenghan Gao, Shizhen Zhang, Xiaofeng Dou, Xingbo Wang, and Quan Li

***Abstract*—Visual Analytics (VA) researchers frequently collaborate closely with domain experts to derive requirements and select appropriate solutions to fulfill these requirements. Despite strides made in exploring requirement and solution spaces, challenges persist due to the absence of guidance in the initial consideration space and the lack of shared problem-solving knowledge, often resulting in suboptimal solutions. To address these issues, we conducted an empirical study of VA research, with a focus on mapping the relations between requirement and solution spaces. Analyzing 220 VA papers, we formulate refined topologies for data, requirements, and solutions. We propose conceptualizing the connections between requirements, data, and solutions through knowledge graphs and utilizing solution paths to encapsulate fundamental problem-solving knowledge in visual analytics research. Through the integration of solution paths into a graph and analyzing their interconnections, we identified a subset of problem-driven design patterns that demonstrated the efficacy of our approach. By externalizing problem-solving knowledge and formulating problem-driven design patterns, our aim is to streamline the exploration of consideration space, facilitating the inclusion of "good" solutions, and establish a benchmark for shared design decisions among researchers and readers.**

***Index Terms*—Visual analytics, design patterns, problem-solving, typology.**

I. INTRODUCTION

VISUAL Analytics (VA) integrates automated analysis techniques with interactive visualizations for effective understanding, reasoning, and decision making based on data [1]. Being an application-oriented discipline, we often derive practical requirements through close collaboration with domain experts. Subsequently, we generate multiple solutions and select a *good* one to meet the requirements [2]. Sedlmair et al. [2] have indicated a common pitfall in this process: considering too few solutions and prematurely committing to a selected one. However, there exists a significant gap between requirements and solutions, leading to a lack of guidance on which options to consider. This gap can result in getting stuck in a "local optimum", as similarly indicated in previous literature [3], [4]. To systematically develop and justify our VA systems, it is crucial to investigate and elaborate on the mapping between requirements and solution spaces.

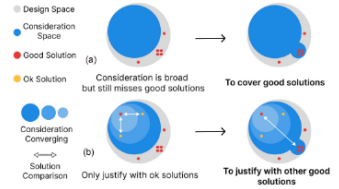

Fig. 1. An ill-positioned consideration space (e.g., with only *bad* and *ok* solutions) results in (a) the selection of *ok* solutions rather than *good* ones, and (b) the justification of current solution with *ok* solutions rather than *good* ones.

Prior research has made strides in refining both requirement and solution spaces of VA systems through investigations into analytical tasks [5], composite visualizations [6], interaction techniques [7], as well as task-specific [8] and data-specific [9] visualizations. However, challenges persist when transitioning from requirements to solutions, emanating from two critical perspectives: *1) Broad consideration can still miss good solutions.* Methodologically, visualization models [2], [10], [11] have been proposed to shape the structure of VA research. These models generally advocate starting with a broad consideration space and progressively narrowing down to a *good* solution guided by design principles, guidelines, and domain expert discussions. However, these models often prioritize variance (i.e., emphasize "broad") over balancing accuracy. The transition from requirements analysis to design specification often relies on intuitions, and design decisions are frequently undocumented [4]. Therefore, even skilled researchers can miss *good* solutions [12] when the initial consideration space lacks guidance [13], [14], even if the consideration space shifts and expands subsequently. As depicted in Fig. 1(a), an inadequately situated consideration space (e.g., containing only *bad* solutions and *ok* solutions) leads

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New approaches for VA system building

VA-Blueprint: Uncovering Building Blocks for Visual Analytics System Design

Leonardo Ferreira, Gustavo Moreira, and Fabio Miranda

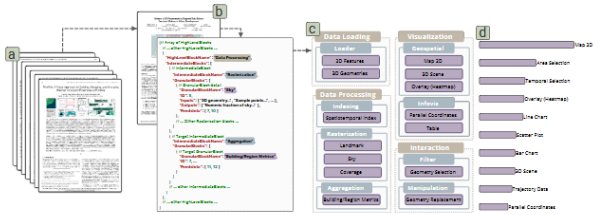

Fig. 1: VA-Blueprint provides a structured way to uncover and organize the fundamental building blocks used in visual analytics (VA) systems. (a) The process begins with the curation of a corpus of VA system research papers. (b) From each paper, core components and their relationships are systematically extracted and structured into a formal, hierarchical blueprint (represented as JSON), detailing the system's components and their dependencies. (c) This blueprint categorizes components across multiple levels of abstraction: high-level stages (e.g., Data Processing), intermediate groups (e.g., Rasterization), and specific granular blocks (e.g., Sky Computation). (d) The final knowledge base comprises 101 machine-readable blueprints encoding component hierarchies, and data and interaction dependencies.

Abstract— Designing and building visual analytics (VA) systems is a complex, iterative process that requires the seamless integration of data processing, analytics capabilities, and visualization techniques. While prior research has extensively examined the social and collaborative aspects of VA system authoring, the practical challenges of developing these systems remain underexplored. As a result, despite the growing number of VA systems, there are only a few structured knowledge bases to guide their design and development. To tackle this gap, we propose VA-Blueprint, a methodology and knowledge base that systematically reviews and categorizes the fundamental building blocks of urban VA systems, a domain particularly rich and representative due to its intricate data and unique problem sets. Applying this methodology to an initial set of 20 systems, we identify and organize their core components into a multi-level structure, forming an initial knowledge base with a structured blueprint for VA system development. To scale this effort, we leverage a large language model to automate the extraction of these components for other 81 papers (completing a corpus of 101 papers), assessing its effectiveness in scaling knowledge base construction. We evaluate our method through interviews with experts and a quantitative analysis of annotation metrics. Our contributions provide a deeper understanding of VA systems' composition and establish a practical foundation to support more structured, reproducible, and efficient system development. VA-Blueprint is available at urbantk.org/va-blueprint.

Index Terms—Visual analytics, large language models, knowledge base, system development, urban visual analytics

1 Introduction

Visual analytics (VA) systems help users make sense of complex data by combining analytics with interactive visualizations. Their usefulness has been demonstrated across diverse domains, including biology [27], healthcare [28], urban planning [17], and climate science [20]. However, building VA systems remains a highly complex and iterative process. System builders must integrate data processing techniques, analytics capabilities, and visualization strategies while ensuring interpretability for domain experts and maintaining efficiency to support interactivity. Despite their significance and inherent challenges, VA system development is often approached in an ad-hoc manner, with minimal reuse of existing components. Given the necessity to address the unique analytical and visualization needs of specific domains, these systems are usually bespoke solutions, leading to fragmented development efforts, making them difficult to extend, adapt, or reuse across projects. This tension, between the need for tailored solutions and the demand for scalable, extensible systems, presents a fundamental challenge in VA system development [12, 58]. On one hand, VA systems must be grounded by domain-specific data, tasks, and analytical workflows: a visualization technique that works well for transportation may be inadequate for urban planning, just as an analytical model for climate science may not be applicable to public health. On the other hand, as VA systems become increasingly complex, the lack of systematic and reusable components hampers the design process and increases the effort needed for development.

Currently, there is a lack of practical knowledge bases that document the building blocks of VA systems. Initial steps have produced valuable taxonomies for visualization tasks and techniques [5]. More recently, new initiatives have emerged to catalog and structure design elements

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A Multimodal Framework for Understanding Collaborative Design Processes

Maurice Koch, Nelusa Pathmanathan, Daniel Weiskopf, and Kuno Kurzhals

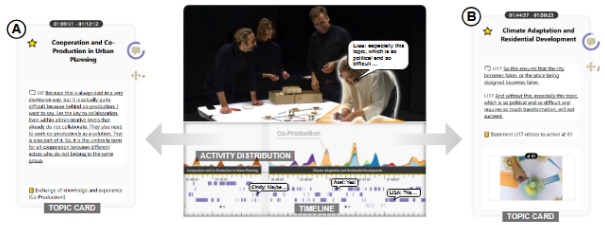

Fig. 1: We record multimodal data from collaborative workshops to derive artifacts from discussions. Streamgraphs show activity and visual attention to different regions of the working area. Activity timelines indicate when different roles in the workshop contributed to a discussion. Editable topic cards are derived automatically from transcripts, serving as elements to build topic summaries. The examples show four participants who discuss the topic of *Co-Production* in urban planning. Lisa is currently active in the part of the working area that concerns Co-Production (●). Topic Cards (A) and (B) provide a multimodal summary of two key events concerning this discussion that the analyst has authored. (*Images edited with Stable Diffusion for privacy.*)

Abstract— An essential task in analyzing collaborative design processes, such as those that are part of workshops in design studies, is identifying design outcomes and understanding how the collaboration between participants formed the results and led to decision-making. However, findings are typically restricted to a consolidated textual form based on notes from interviews or observations. A challenge arises from integrating different sources of observations, leading to large amounts and heterogeneity of collected data. To address this challenge we propose a practical, modular, and adaptable framework of workshop setup, multimodal data acquisition, AI-based artifact extraction, and visual analysis. Our interactive visual analysis system, reCAPit, allows the flexible combination of different modalities, including video, audio, notes, or gaze, to analyze and communicate important workshop findings. A multimodal streamgraph displays activity and attention in the working area, temporally aligned topic cards summarize participants' discussions, and drill-down techniques allow inspecting raw data of included sources. As part of our research, we conducted six workshops across different themes ranging from social science research on urban planning to a design study on band-practice visualization. The latter two are examined in detail and described as case studies. Further, we present considerations for planning workshops and challenges that we derive from our own experience and the interviews we conducted with workshop experts. Our research extends existing methodology of collaborative design workshops by promoting data-rich acquisition of multimodal observations, combined AI-based extraction and interactive visual analysis, and transparent dissemination of results.

Index Terms—Collaborative workshops, multimodal analysis of design processes, combination of AI and interactive visualization, design study methodology, eye tracking, reCAPit.

1 Introduction

Design processes in the field of visualization, as well as other domains, are often part of a collaborative effort, comprising multiple stages from acquiring knowledge about potential means to solve a problem to the implementation and analysis of a new design [45]. One way to address the earlier stages of the design process is conducting creative visualization-opportunities (CVO) workshops, which typically involve domain and visualization expertise to address the problem at hand [25].

During such workshops, participants create fragments (e.g., physical objects, utterances, sketches) as outcomes that foster discussion and potentially support decision-making. These fragments further serve as a means to document the whole process by textual summarization of findings, images, or objects. If researchers want to communicate how these results were derived, a temporal analysis of the workshop is required, by someone taking notes throughout or by recording audio and/or video, leading to an increased analysis effort.

Workshop analysis often relies on open coding [12] of recorded audio and handwritten notes, which is time-consuming and requires extensive analysis of the material. The high workload associated with established workflows discourages workshop conductors from recording or harnessing material from other sources, such as video or physiological sensors. However, video has the potential of capturing activities that are not accompanied by verbal utterances, such as gestures, activities, and movement. Many aspects of the video can be analyzed automatically

• *All authors are with Visualization Research Center (VISUS), University of Stuttgart. E-mail: {first name}.{last name}@visus.uni-stuttgart.de.*

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Qualitative Study for LLM-assisted Design Study Process: Strategies, Challenges, and Roles

Shaolun Ruan, Rui Sheng, Xiaolin Wen, Jiachen Wang, Tianyi Zhang, Yong Wang, Tim Dwyer, and Jiannan Li

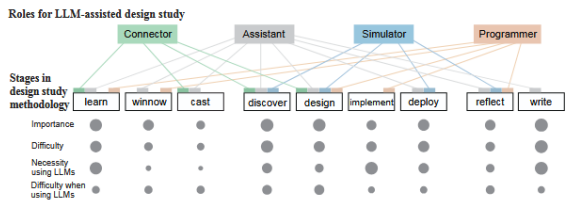

Fig. 1: An overview of the summarized roles of LLMs in the LLM-assisted design study process. Following the framework proposed by Sedlmair et al. [34], we coded all nine stages of the design study by the compiled roles: 1) *Connector:* LLMs are used to bridge gaps between visualization researchers and domain experts; 2) *Assistant:* LLMs are commonly used for repetitive tasks to enhance productivity; 3) *Simulator:* LLMs support simulating user behavior, predicting feedback, and testing workflows to uncover usability issues and hidden use cases; 4) *Programmer:* LLMs are asked to assist in coding tasks, from prototyping to debugging. The bottom section shows ratings from 30 researchers on the importance, difficulty, and necessity of using LLMs, as well as the difficulty when using LLMs at each stage.

Abstract—Design studies aim to develop visualization solutions for real-world problems across various application domains. Recently, the emergence of large language models (LLMs) has introduced new opportunities to enhance the design study process, providing capabilities such as creative problem-solving, data handling, and insightful analysis. However, despite their growing popularity, there remains a lack of systematic understanding of how LLMs can effectively assist researchers in visualization-specific design studies. In this paper, we conducted a multi-stage qualitative study to fill this gap, which involved 30 design study researchers from diverse backgrounds and expertise levels. Through in-depth interviews and carefully-designed questionnaires, we investigated strategies for utilizing LLMs, the challenges encountered, and the practices used to overcome them. We further compiled the roles that LLMs can play across different stages of the design study process. Our findings highlight practical implications to inform visualization practitioners, and also provide a framework for leveraging LLMs to facilitate the design study process in visualization research.

Index Terms—Design Study, Large Language Models (LLMs), Qualitative Study, Visualization

1 Introduction

Sedlmair *et al.* introduced the term *design study* to describe an applied research methodology that focuses on creating visualizations to address specific, real-world problems [34]. More specifically, they defined a visualization design study as "a project in which visualization researchers analyze a specific problem faced by domain experts and design a visualization system that supports solving this problem" [34]. Building upon this definition, a nine-stage methodology framework was proposed for conducting design studies, as shown in Fig. 1, which has become a common guidance for this field of research in the visualization community. Design studies are crucial because they bridge the gap between theoretical visualization research and practical applications, ensuring that visualization tools are both usable and useful in real-world contexts [29]. Basically, researchers conduct design studies through labor-intensive processes (*e.g.*, connecting with domain experts, extracting requirements via interviews, and iteratively designing and developing visualization systems), which require significant effort and refined knowledge to gain effective solutions.

Meanwhile, Large Language Models (LLMs) have shown increasing

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New approaches for VA system building

Understand

- Learn about the target domain and the practices, needs, problems, and requirements of the domain expert. [Sedlmair et al., 2012]
- Understand the problem, domain, and target users. Gather information to find user needs. [McKenna et al., 2014]
- Undertake domain-specific data analysis, analyze data collaboratively with domain experts. [Hell et al., 2019]
- Elicitation of tasks from stakeholders. [Oppermann and Munzner, 2020]

Ideate

- Early designs to illustrate the essence of an idea. [Lloyd and Dykes, 2011]
- Digital interactive prototypes. [Lloyd and Dykes, 2011]
- Generation and validation of data abstractions, visual encodings, and interaction mechanisms. [Sedlmair et al., 2012]

Make

- Implementation of the tool. [Lloyd and Dykes, 2011]
- Implementation of software prototypes and tools. [Sedlmair et al., 2012]
- Concretize ideas into tangible prototypes. [McKenna et al., 2014]
- Implementation of the software and tools. [Oppermann and Munzner, 2020]

Evaluate

- See previous weeks!

Understand: Problem → Tasks → Requirements

- Goal: Build a shared understand of the domain + users + data + tasks + constraints, and decide what "success" of a VA system means.
- Pain points:
  - Labor intensive, with interviews, observations, workshops, but it's where most of the downstream success is determined.

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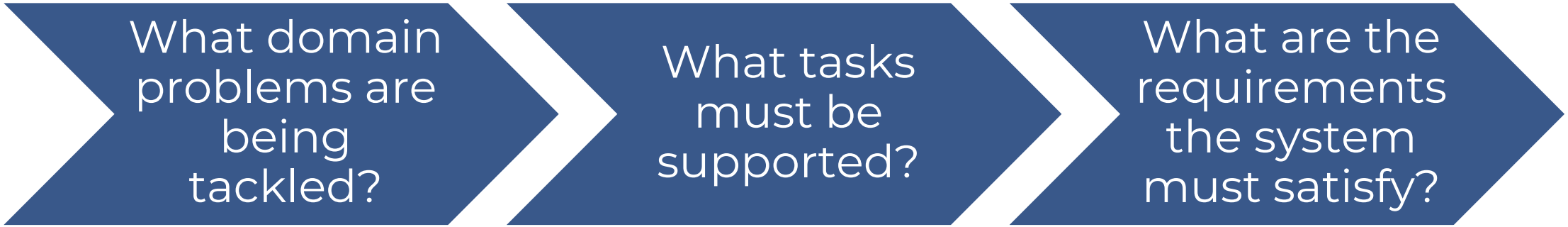


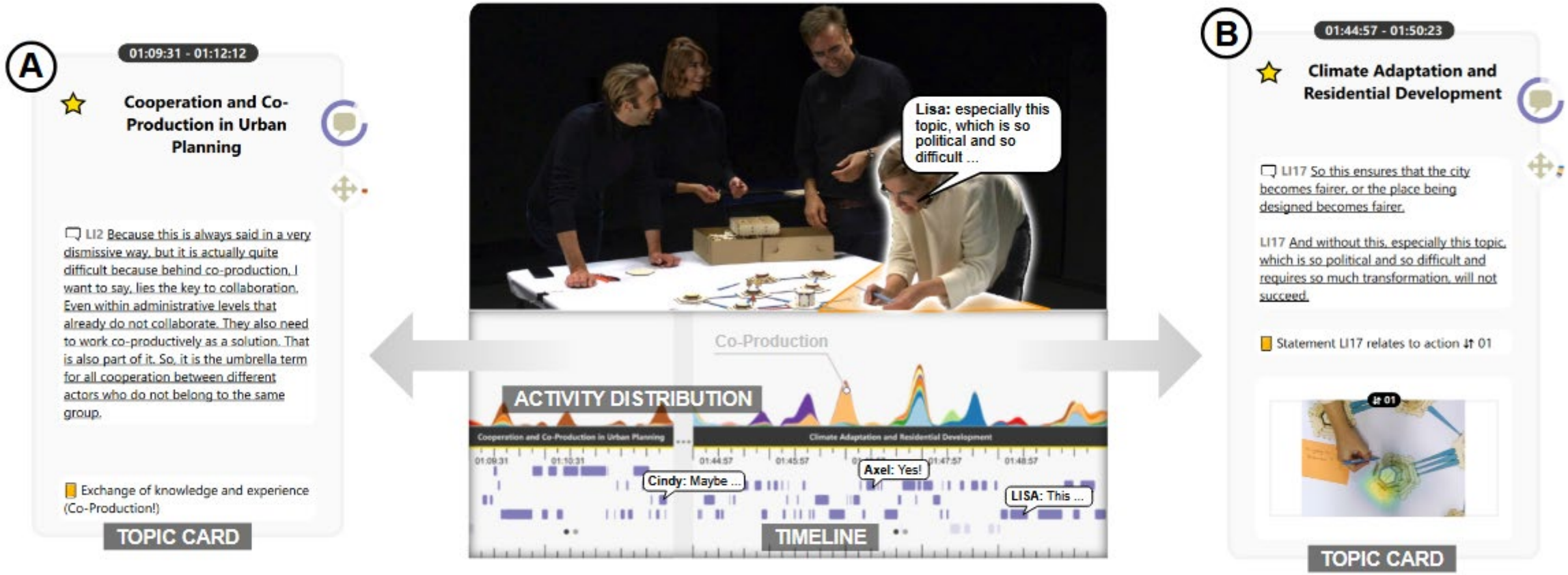
Understand: Problem → Tasks → Requirements

- **Problem framing:** What domain problems are being tackled?
  - A VA system design starts by grounding the work in a real problem faced by domain experts.
  - What is in scope / out of scope?
  - Why this step is hard in VA system building:
    - VA systems are inherently complex + iterative to build and must combine several components while staying interactive.
  - **AI opportunities:**
    - **Faster domain ramp-up:**

      LLMs help fill “gaps in domain-specific terminology and foundational concepts” (Ruan et al., 2026)
    - **Faster domain translation:**

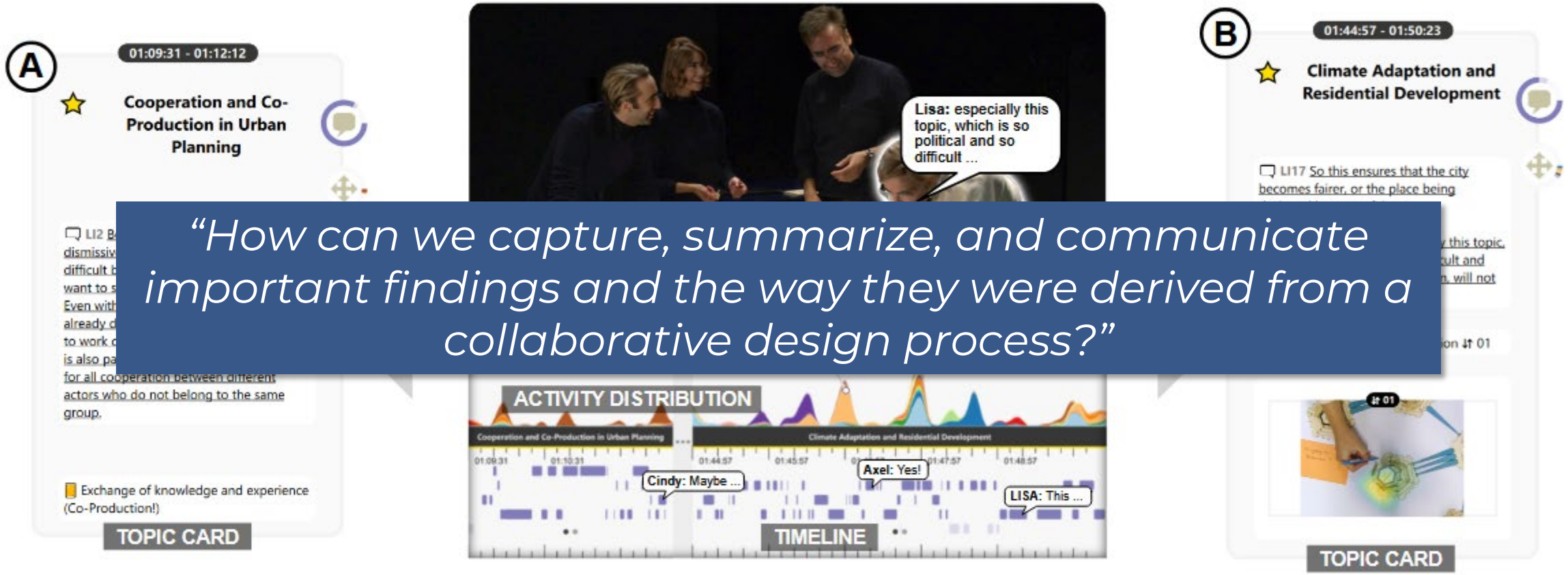
      LLMs can “assist in translating domain-specific knowledge into actionable design insights, decomposing requirements into sub-tasks, and providing design inspiration” (Ruan et al., 2026)

Understand: Problem → Tasks → Requirements



Koch et al., 2026

Understand: Problem → Tasks → Requirements



"How can we capture, summarize, and communicate important findings and the way they were derived from a collaborative design process?"

Koch et al., 2026

Understand: Problem → Tasks → Requirements

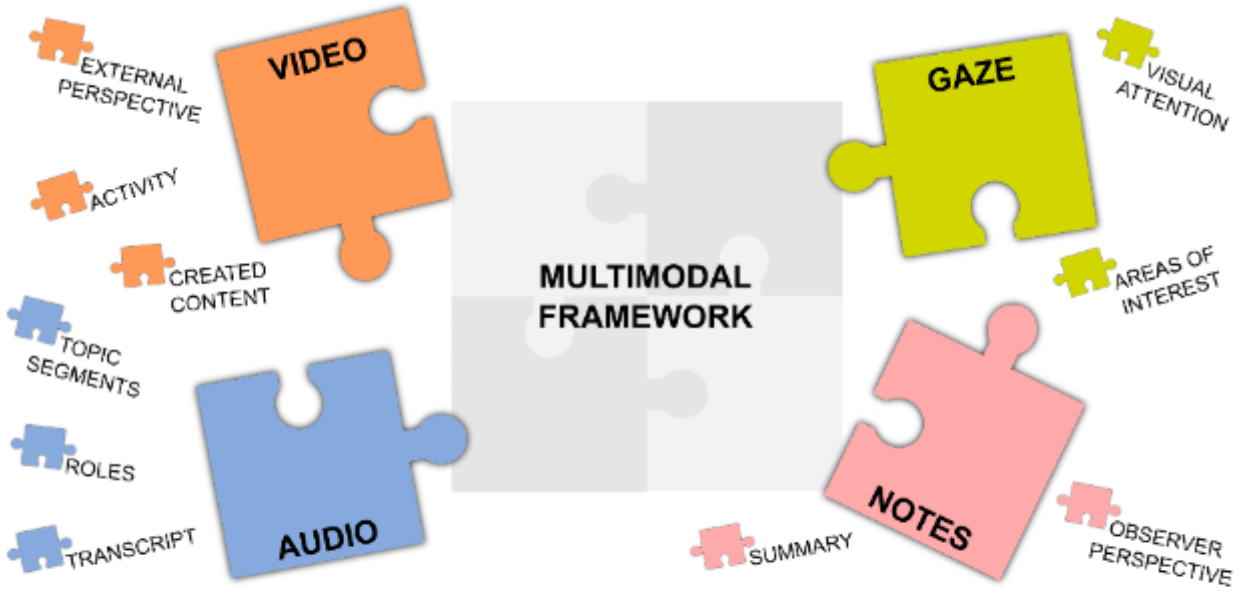


Fig. 2: Our framework supports the flexible combination of recording audio, video, notes, and/or gaze for a multimodal view on collaborative workshop processes. From this data, we extract artifacts for analysis.

Understand: Problem → Tasks → Requirements

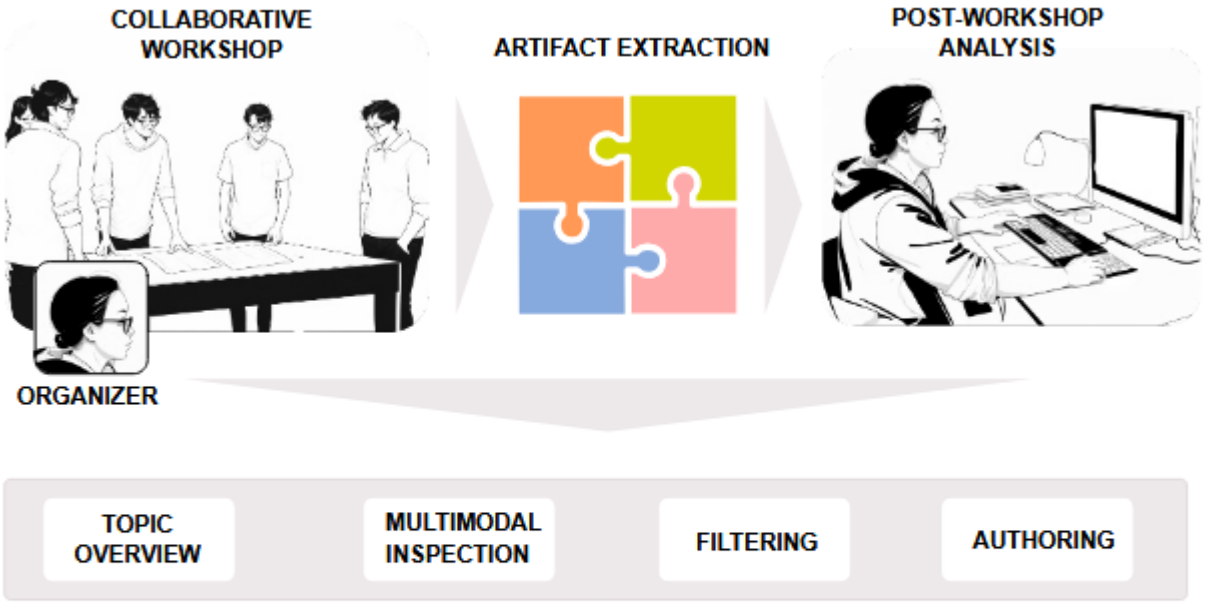


Fig. 4: Data is captured during a workshop and processed to extract artifacts. A post-experiment analysis is conducted by workshop organizers. The analysis comprises multiple methods to inspect and author artifacts to derive summaries for sensemaking and dissemination. (*Images edited with Stable Diffusion for privacy.*)

- Multimodal framework for workshop conductors to collect, consolidate, and analyze resulting multimodal data.
- Researchers can revisit important events from different perspectives of the multimodal data.
- Overview of all data streams, and drill-down into each individual stream.

Understand: Problem → Tasks → Requirements

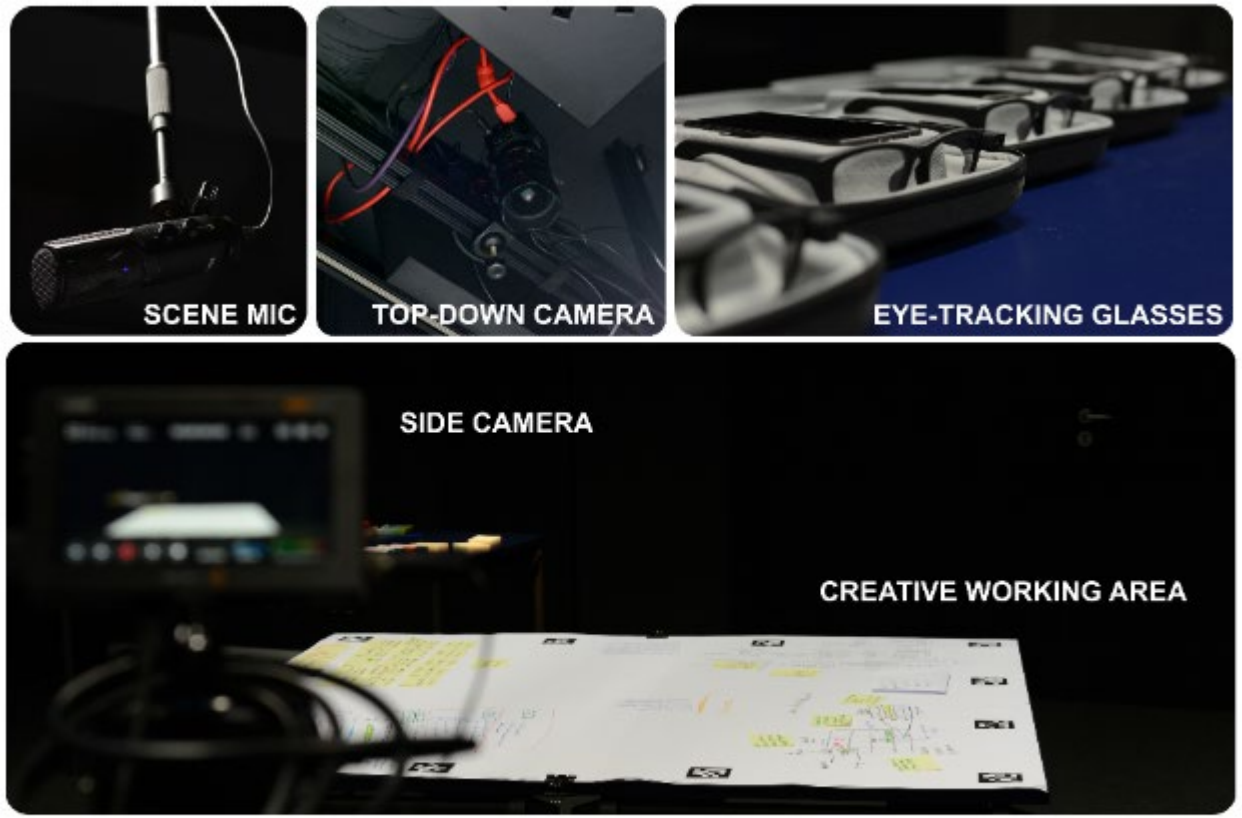


Fig. 3: Workshop setup with one side camera and one top-down camera. Audio was recorded with a scene microphone mounted above the work area, gaze with multiple eye-tracking glasses.

- Multimodal framework for workshop conductors to collect, consolidate, and analyze resulting multimodal data.
- Researchers can revisit important events from different perspectives of the multimodal data.
- Overview of all data streams, and drill-down into each individual stream.

Understand: Problem → Tasks → Requirements

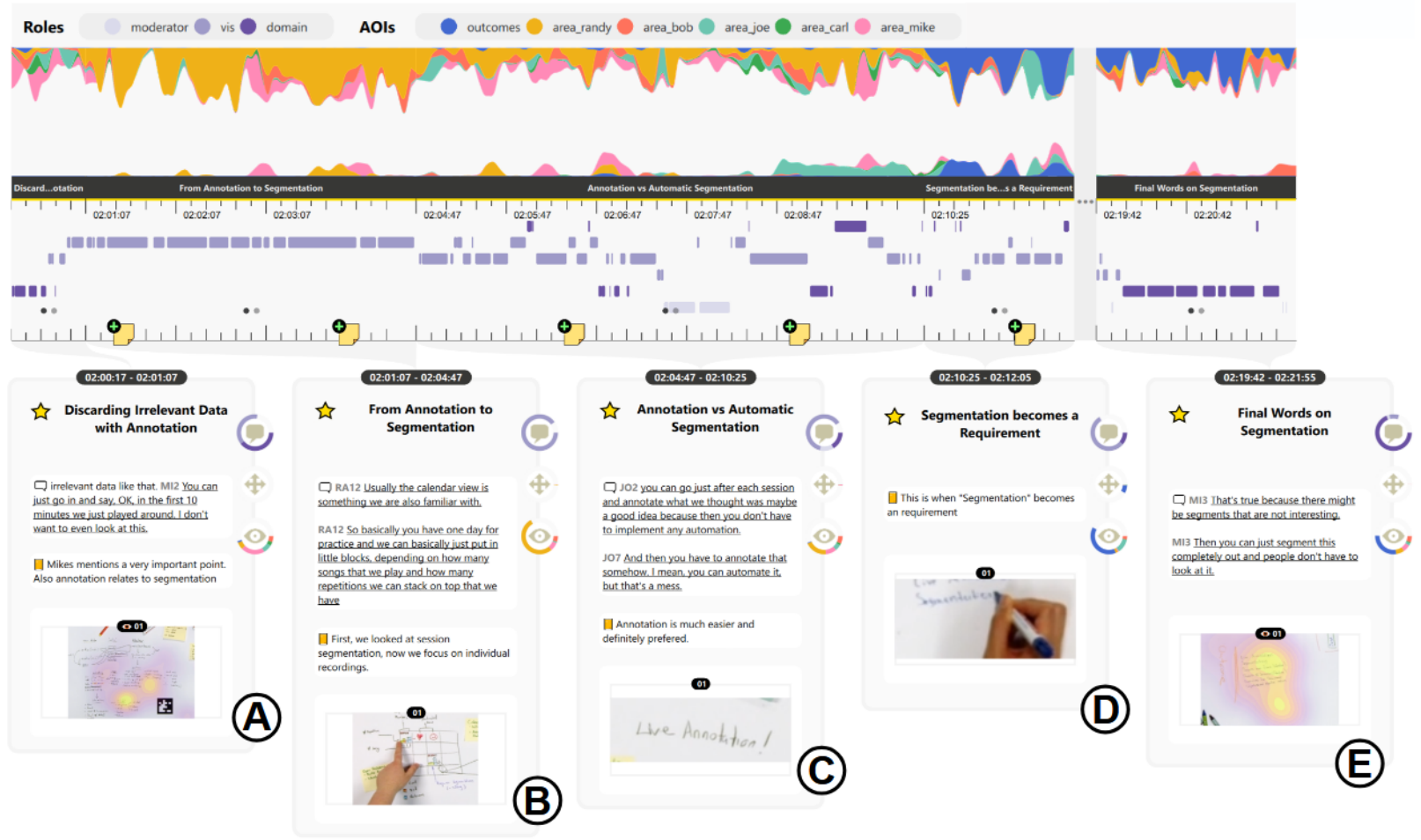


Fig. 9: Final analysis state related to the requirement *Segmentation* comprising the five topic segments (A)–(E). The streamgraphs show activity in Randy's area as he explains his visualization and adds the phrase "Live Annotation!" to his sketches.

Understand: Problem → Tasks → Requirements

- **Task elicitation:** What operations performed by the user must be supported?
- **Requirement gathering:** What are the necessary functionalities, capabilities, and characteristics of a system?

Tasks:
Lower level and more focused on specific actions that users need

Requirements:
Oriented towards reflecting user intentions in terms of system characteristics.

Understand: Problem → Tasks → Requirements

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- **Requirement gathering:** What are the necessary functionalities, capabilities, and characteristics of a system?

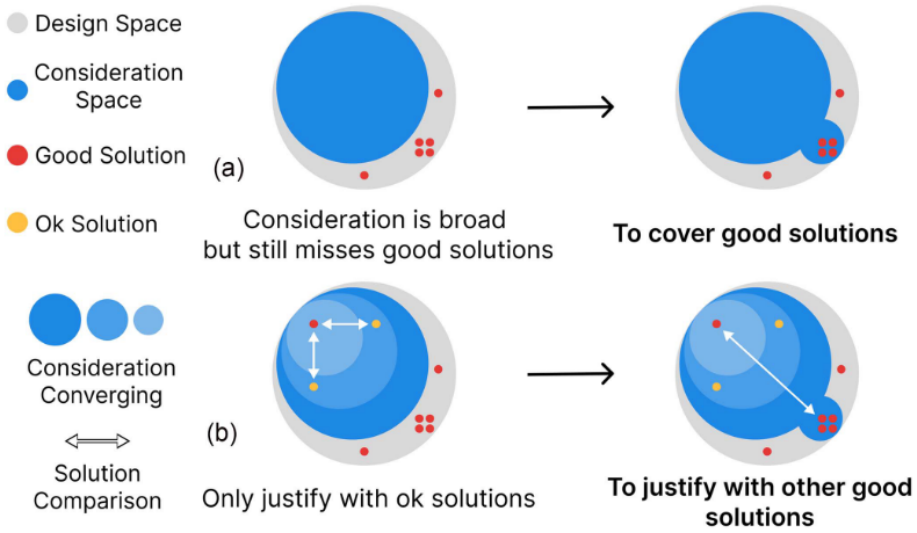

Fig. 1. An ill-positioned consideration space (e.g., with only *bad* and *ok* solutions) results in (a) the selection of *ok* solutions rather than *good* ones, and (b) the justification of current solution with *ok* solutions rather than *good* ones.

Wu et al., 2026

Understand: Problem → Tasks → Requirements

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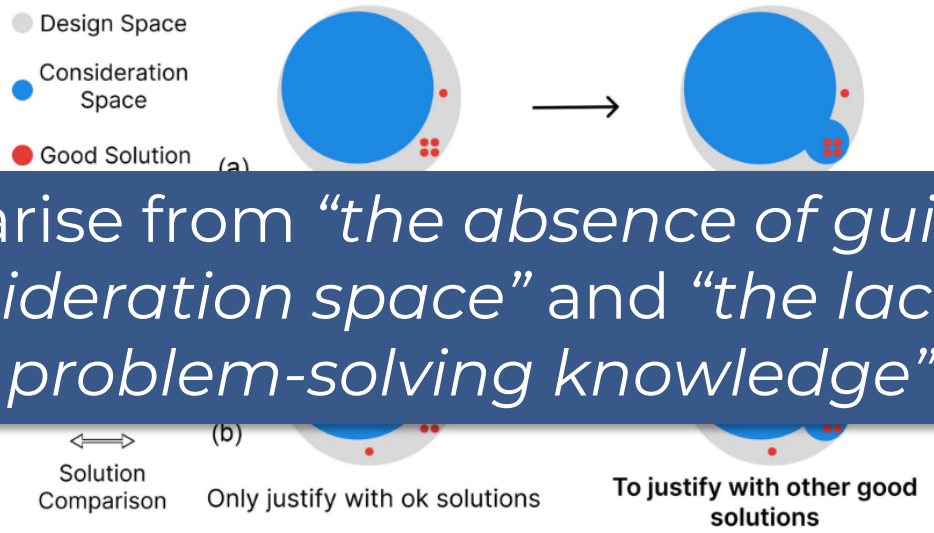

Challenges arise from *"the absence of guidance in the initial consideration space"* and *"the lack of shared problem-solving knowledge"*

Fig. 1. An ill-positioned consideration space (e.g., with only *bad* and *ok* solutions) results in (a) the selection of *ok* solutions rather than *good* ones, and (b) the justification of current solution with *ok* solutions rather than *good* ones.

Wu et al., 2026

Understand: Problem → Tasks → Requirements

- Extract “material units” from each VA paper:
  - Segment each paper into discrete units describing requirements, data, and solutions.
- Code each unit using shared typologies.
  - Requirements
  - Data
  - Solutions
- Represent each paper as a knowledge graph and derive “solution paths”
  - **Requirement → Data** edges (“is imposed on”)
  - **Data → Solution** edges (“is utilized by”)
  - **Solution → Solution** edges (“as basis of”)

Understand: Problem → Tasks → Requirements

TABLE III
THE SOLUTION TYPOLOGY DERIVED FROM CODING SOLUTIONS

A) Data Manipulation				
Eff\Tar	*Augment*	*Reduce*	*Search*	*Modify*
Item	Real-Time Input	Excluding	Retrieval	Rectification
	User Input	Sampling	Similarity Calculation	Parameter Tuning
Dimension	Algorithmic Calculation	Dimensionality Reduction		
	Modeling	Feature Selection	Explainability	Wrangling
	Clustering & Grouping			

B) Visualization							
Composite							
Juxtaposition			*Overlay*				*Nesting*
Similar		*Unsimilar*	*Coordinate system-related*		*Coordinate system-unrelated*		Nesting
Nonsymmtrical	*Symmtrical*	*Continuous*	*Sharing*	*Providing*	*With connections*	*W/O connections*	
Repetition	Mirror	Stack	Co-axis	Coordinate	Annotation	Large panel	
Non-composite							
With layout					*Without Layout*		
Aligned		Flexible			Basics		

C) Interaction										
Filtering	Selecting	Abstract/ Elaborate	Overview and Explore	Connect/ Relate	Reconfigure	Encode	History	Extraction of Features	Participantion/ Collaboration	Gamification

The typology combines previous typologies on visualization [6] and interaction [7], while introducing a new solution type of a) data manipulation. The data manipulations are organized by two axes: *effect* (horizontal) ranges from augment to modify and denotes the expected outcomes when applying the manipulation; *target* (Vertical) denotes the entities on which the manipulation operates.

Wu et al., 2026

Understand: Problem → Tasks → Requirements

A **solution path** is then a path from a requirement node to a "leaf" solution node (zero outdegree), meant to capture one atomic, coherent way that paper addresses the requirement.

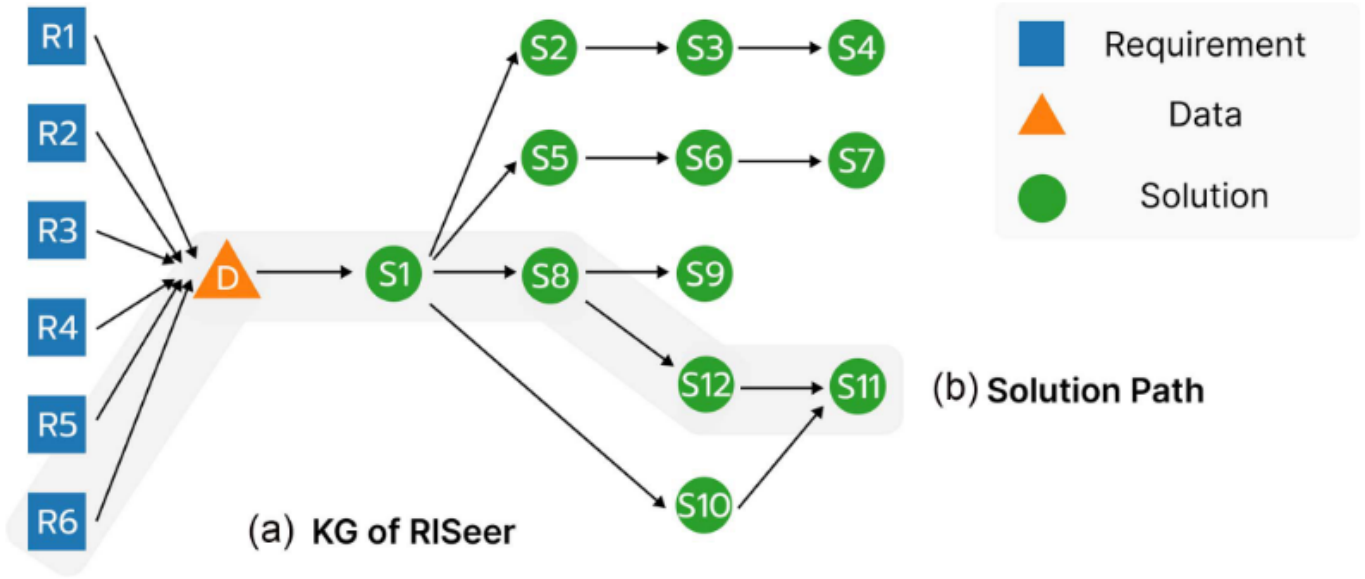

Fig. 6. We represent the problem-solving knowledge in VA research as a knowledge graph of requirements, data, and solutions. (a) An example of KG of RISeer; (b) An instance of solution path.

Understand: Problem → Tasks → Requirements

How it can be used in practice?

1. "Design search" for requirement→ candidate solution strategies
   - Given a requirement (and data), we can traverse the base to see what solution components and sequences have been used in prior VA work.
   - Wu et al. argue that rather than a single recommendation, their work can support a **panoramic view** of the landscape of VA solutions.
2. Evidence for design justification
   - Used as evidence, showing how requirements have been addressed through different components.
3. Deriving reusable "problem-driven design patterns"
   - Frequently occurring subgraphs / paths are treated **as design patterns**. A pattern library where patterns are anchored in many instances from literature.
4. Tooling: a knowledge base for teams
   - A graph database that supports queries such as "What interaction types frequently appear downstream of this visualization?" or "What are the sparse areas (few paths) suggesting research gaps?"

Ideate: Data exploration → Sketching → Prototyping

- Goal: Produce multiple plausible solution directions.
- Pain points:
  - The space of possible solutions is huge.
  - Early design gets anchored to whatever you explore first (local optimum).
  - Iteration trash: ideas bounce around without tracking what changed and why; teams repeat mistakes.
- **AI opportunities:**
  - **Idea generation:**
    Propose alternative visualizations.
  - **Rapid data profiling:**
    Auto-summarize dataset types, quality issues, etc.
  - **Visualization iteration:**
    Provide feedback for visualizations:
  - **Summarization:**
    Summarize notes and transcripts into records of options and decisions made.

Ideate: Data exploration → Sketching → Prototyping

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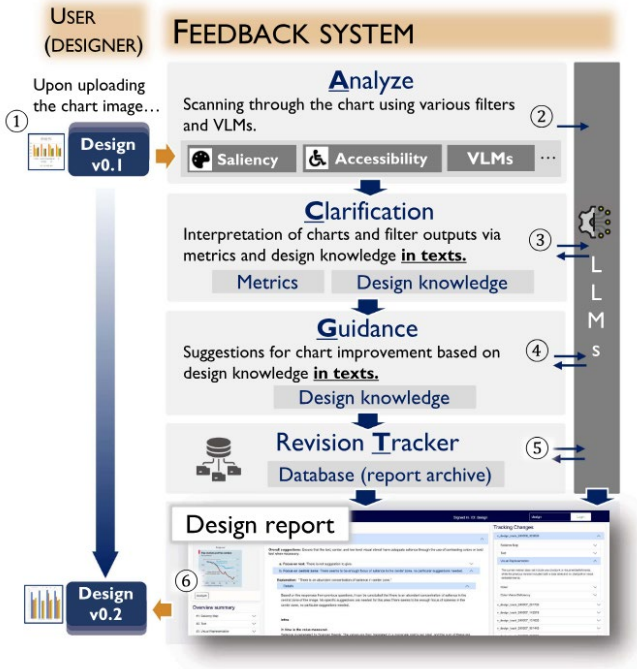
Ideate: Data exploration → Sketching → Prototyping

- Shin et al. propose Visualizationary, an LLM- and VLM-based tool to support visualization iteration.
- Design knowledge is incorporated as preambles in the prompt.

$$T_{acg} = [Q] +$$
"Solve the problem based on this guideline:" +
[cond] + [filter-suggestions].
$$T_t = [Q] +$$
"Here are details about the current version:" +
[curr-output] + [curr-interpretations]
+ "Here are details about the previous version:" +
[prev-output] + [prev-interpretations].

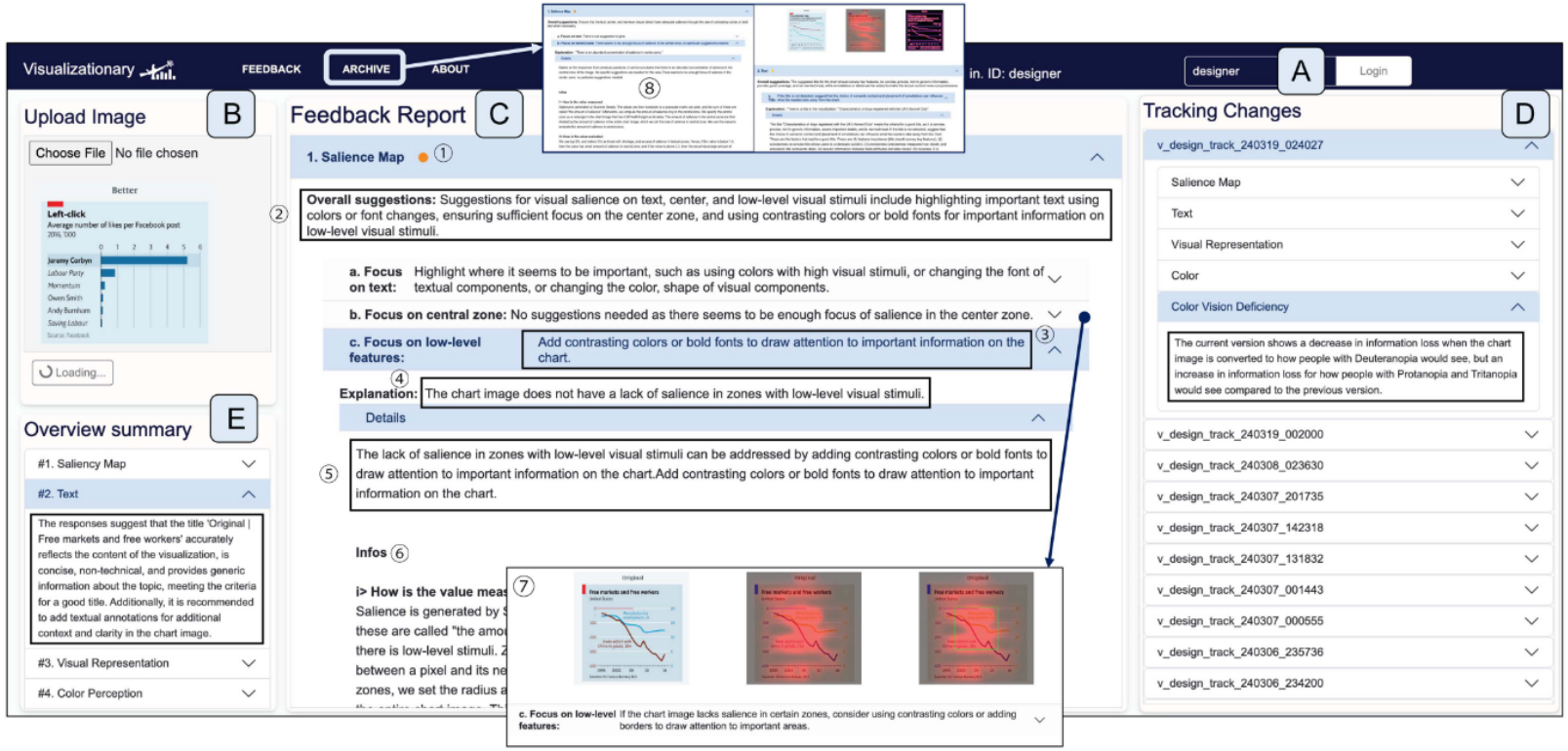


Shin et al., 2026

Ideate: Data exploration → Sketching → Prototyping

- **Q:** Analyze the visual salience on text. Provide interpretations in 2 sentences.
- **CONDITIONS:** If no visual salience is detected, say: "No salience is detected in the chart image.“
- **FILTER-SUGGESTIONS** (design knowledge, optional): If textual elements lack salience, then increase contrast, increase font weight/size, or reduce competing salience from decorative elements.

Ideate: Data exploration → Sketching → Prototyping



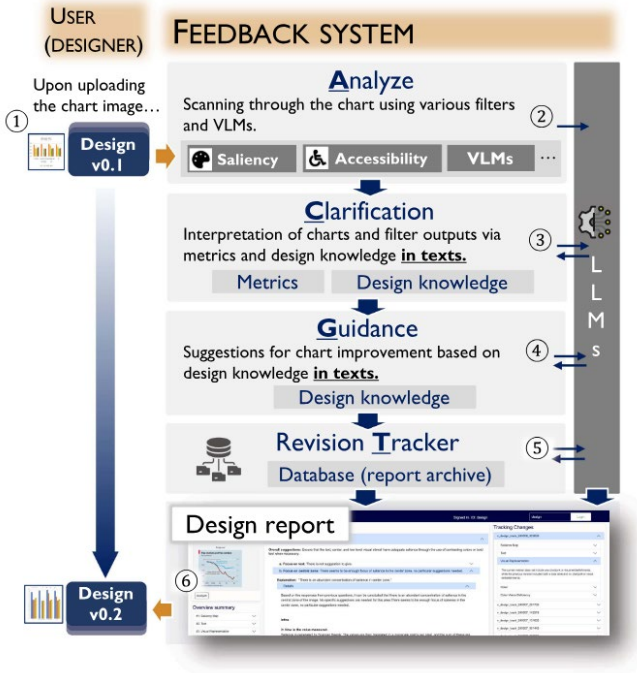


Fig. 2. Visualizationary interface. Example of an interactive report on a visualization artifact generated using our novel analyze-clarify-guide-track (ACGT) framework. The image is a depiction of our system, Visualizationary where a new chart image (Fig. 1(B)) is being updated while the current report is about the past visualization image. The report is generated by automatically analyzing ("analyze") the artifact using an expandable collection of image filters grouped into categories: salience, text, visual representation, color, and accessibility. After reporting results from each image filter (center column), the report then interprets ("clarify") the findings using natural language understandable to a novice visualization designer. This is followed by natural language suggestions ("guide") on how to address weaknesses in the design. Finally, the right side of the report shows changes over time ("track") as the designer iteratively refines the artifact.

Shin et al., 2026

Make: Implementation → Integration → Iteration

- Goal: Turn sketches and prototypes into a working VA system.
- Pain points:
  - Architecture drift: implemented system diverges from the intended design.
  - Integration complexity: wiring components is complex.
  - Regression risk: every change breaks something else.
- **AI opportunities:**
  - **Code scaffolding:**
    Generate and enforce consistent patterns.
  - **Data pipeline assistance:**
    Suggest transformations, schemas.
  - **Documentation as-you-build:**
    Auto-generate API docs, onboarding / tutorial content.

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Make: Implementation → Integration → Iteration

LightVA: Lightweight Visual Analytics With LLM Agent-Based Task Planning and Execution

Yuheng Zhao, Junjie Wang, Linbing Xiang, Xiaowen Zhang, Zifei Guo, Cagatay Turkay, Yu Zhang, and Siming Chen

***Abstract*—Visual analytics (VA) requires analysts to iteratively propose analysis tasks based on observations and execute tasks by creating visualizations and interactive exploration to gain insights. This process demands skills in programming, data processing, and visualization tools, highlighting the need for a more intelligent, streamlined VA approach. Large language models (LLMs) have recently been developed as agents to handle various tasks with dynamic planning and tool-using capabilities, offering the potential to enhance the efficiency and versatility of VA. We propose LightVA, a lightweight VA framework that supports task decomposition, data analysis, and interactive exploration through human-agent collaboration. Our method is designed to help users progressively translate high-level analytical goals into low-level tasks, producing visualizations and deriving insights. Specifically, we introduce an LLM agent-based task planning and execution strategy, employing a recursive process involving a planner, executor, and controller. The planner is responsible for recommending and decomposing tasks, the executor handles task execution, including data analysis, visualization generation and multi-view composition, and the controller coordinates the interaction between the planner and executor. Building on the framework, we develop a system with a hybrid user interface that includes a task flow diagram for monitoring and managing the task planning process, a visualization panel for interactive data exploration, and a chat view for guiding the model through natural language instructions. We examine the effectiveness of our method through a usage scenario and an expert study.**

***Index Terms*—Visual analytics, task planning, large language model agent, mixed-initiative interaction.**

I. Introduction

VISUAL analytics (VA) deciphers complex datasets with data mining and interactive visualizations [1], [2]. However, building and using a VA system can be a costly endeavor that encompasses several main stages: goal understanding, task decomposition, data modeling, and visualization creation to discover insights. A key challenge is that this process is iterative, requiring continual refinement based on evolving needs [3]. Different tasks necessitate various data analysis and visualization methods to form a VA system. While using the system, tasks may evolve based on the insights gained, necessitating ongoing iterations until the analytical goals are achieved [4]. Consider a scenario where the goal is to identify high-risk events from social media data. Users may first need an overview from different perspectives, such as the distribution of keywords over time or changes in sentiment for risk analysis. If outliers are identified, users may need further details, such as spatial distribution or entity relationships. In this process, the task space is broad and fluid, requiring efficient task planning and method implementation.

Recent research focuses on data-driven or natural language-based visual data exploration, with an emphasis on automatic visualization generation [5], [6] or insight mining [7], [8]. Large Language Models (LLMs) present a potential for data analysis, supporting dynamic task planning and lower development costs across various scenarios. The reasoning abilities that enable autonomous planning and execution of analytical tasks [9], [10], [11], while code generation capability support creating insightful visualizations efficiently [12], [13], [14]. Additionally, their broad knowledge base makes LLMs versatile tools capable of adapting to diverse data analysis contexts [15], [16]. LEVA [17] integrates LLMs in VA systems to recommend insights for a given task but still cannot generate visualization and data modeling methods adapted to tasks. There is a lack of approaches supporting task planning, VA methods implementation, and interactive analysis with human-agent collaboration.

This paper introduces *LightVA*, a lightweight VA framework with agent-based task planning. The term "lightweight" refers to the framework's focus on reducing the cost of development and using VA systems. Using LLM agents to aid the task planning and execution process. The framework builds upon multi-level relationships, translating high-level goals to low-level tasks and deriving insights through data mining and interactive visualizations. Specifically, the framework employs a recursive process that includes a planner, executor, and controller, which dynamically accommodates task complexity. The planner is responsible for task decomposition, the executor handles task execution, including visualization generation and data analysis, and the controller orchestrates the executor and planner and manages whether tasks continue to be decomposed. We develop a system based on the framework that provides a chat view to support

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Digital Object Identifier 10.1109/TVCG.2024.3496112

VA-Blueprint: Uncovering Building Blocks for Visual Analytics System Design

Leonardo Ferreira, Gustavo Moreira, and Fabio Miranda

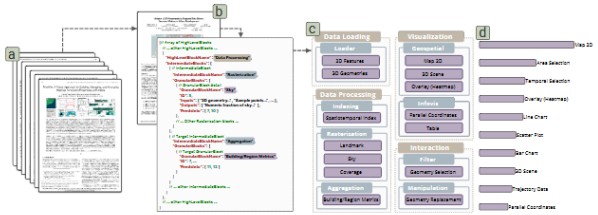

Fig. 1: VA-Blueprint provides a structured way to uncover and organize the fundamental building blocks used in visual analytics (VA) systems. (a) The process begins with the curation of a corpus of VA system research papers. (b) From each paper, core components and their relationships are systematically extracted and structured into a formal, hierarchical blueprint (represented as JSON), detailing the system's components and their dependencies. (c) This blueprint categorizes components across multiple levels of abstraction: high-level stages (e.g., Data Processing), intermediate groups (e.g., Rasterization), and specific granular blocks (e.g., Sky Computation). (d) The final knowledge base comprises 101 machine-readable blueprints encoding component hierarchies, and data and interaction dependencies.

Abstract— Designing and building visual analytics (VA) systems is a complex, iterative process that requires the seamless integration of data processing, analytics capabilities, and visualization techniques. While prior research has extensively examined the social and collaborative aspects of VA system authoring, the practical challenges of developing these systems remain underexplored. As a result, despite the growing number of VA systems, there are only a few structured knowledge bases to guide their design and development. To tackle this gap, we propose VA-Blueprint, a methodology and knowledge base that systematically reviews and categorizes the fundamental building blocks of urban VA systems, a domain particularly rich and representative due to its intricate data and unique problem sets. Applying this methodology to an initial set of 20 systems, we identify and organize their core components into a multi-level structure, forming an initial knowledge base with a structured blueprint for VA system development. To scale this effort, we leverage a large language model to automate the extraction of these components for other 81 papers (completing a corpus of 101 papers), assessing its effectiveness in scaling knowledge base construction. We evaluate our method through interviews with experts and a quantitative analysis of annotation metrics. Our contributions provide a deeper understanding of VA systems' composition and establish a practical foundation to support more structured, reproducible, and efficient system development. VA-Blueprint is available at urbantk.org/va-blueprint.

Index Terms—Visual analytics, large language models, knowledge base, system development, urban visual analytics

1 Introduction

Visual analytics (VA) systems help users make sense of complex data by combining analytics with interactive visualizations. Their usefulness has been demonstrated across diverse domains, including biology [27], healthcare [28], urban planning [17], and climate science [20]. However, building VA systems remains a highly complex and iterative process. System builders must integrate data processing techniques, analytics capabilities, and visualization strategies while ensuring interpretability for domain experts and maintaining efficiency to support interactivity. Despite their significance and inherent challenges, VA system development is often approached in an ad-hoc manner, with minimal reuse of existing components. Given the necessity to address the unique analytical and visualization needs of specific domains, these systems are usually bespoke solutions, leading to fragmented development efforts, making them difficult to extend, adapt, or reuse across projects. This tension, between the need for tailored solutions and the demand for scalable, extensible systems, presents a fundamental challenge in VA system development [12, 58]. On one hand, VA systems must be grounded by domain-specific data, tasks, and analytical workflows: a visualization technique that works well for transportation may be inadequate for urban planning, just as an analytical model for climate science may not be applicable to public health. On the other hand, as VA systems become increasingly complex, the lack of systematic and reusable components hampers the design process and increases the effort needed for development.

Currently, there is a lack of practical knowledge bases that document the building blocks of VA systems. Initial steps have produced valuable taxonomies for visualization tasks and techniques [5]. More recently, new initiatives have emerged to catalog and structure design elements

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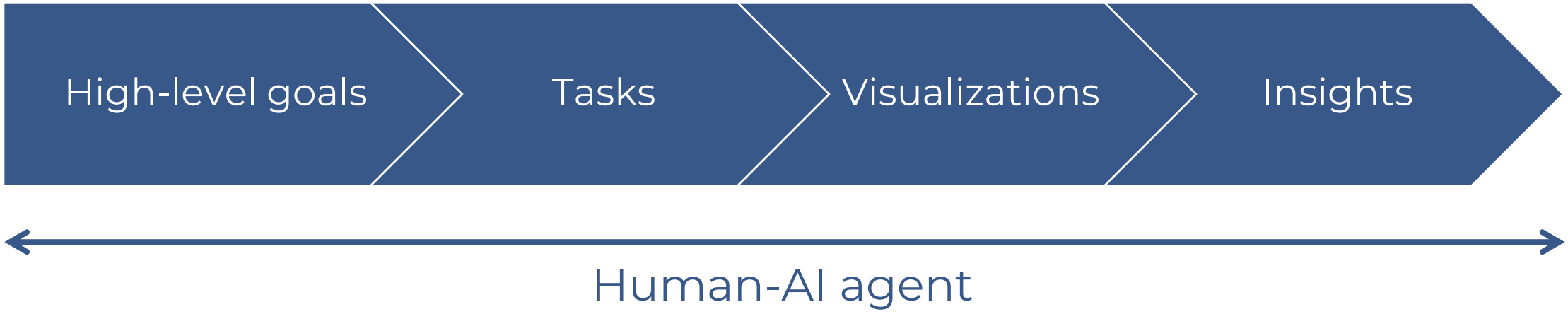
LightVA

- Building and using a VA system can be a costly endeavor.
  - Real barrier: requires programming, data processing, and vis expertise.
- Encompasses several stages:
  1. goal understanding
  2. task decomposition
  3. data modeling
  4. visualization creation
- This process is iterative, requiring continual refinement based on evolving needs.

LightVA

- AI agents suggest a new path forward: **planning + tool use to streamline VA**
- LightVA: a lightweight VA that unifies:

system construction + system use



LightVA: Challenges

- **C1**: Accomodating diverse analysis requirements
  - Data analysts often face challenges of navigating a vast exploration space.
  - Forming hypotheses is complex.
  - They have to figure out the connections between tasks and insights in their mind and propose tasks in the next few steps until achieve the goal.
- **C2**: Developing visualization and data mining tools is time consuming
  - Analysts often lack proficiency in developing and managing VA systems.
  - When tasks involve multiple visualizations, these must be integrated into a linked view.
  - This process requires substantial knowledge of visualization principles and coding skills.

LightVA: Design requirements

- **R1:** Adaptive task planning: tasks should adapt to goals, data, and evolving results (C1)
- **R2:** Flexible visualization generation: data identification/transform + viz choice + interactive views (C2)
- **R3:** Automatic insight generation: complete code-based analyses + highlight key findings (C2)
- **R4:** Multiple view composition: keep prior results, allow merging views + reduce cognitive load (C2)
- **R5:** Intuitive analysis process: make task–viz–insight–progress relationships legible; support human–agent interaction (C1, C2)

LightVA: Conceptual framework

- "Lightweight": **lower expectations** for both *building* VA systems and *using* them
- Key primitives:
  - Goal: high-level purpose statement
  - Task: executable aspect of goal, defined by attributes (type, variables, method, progress)
- **Agent:** Task planning and execution
- **Human:** Refinement

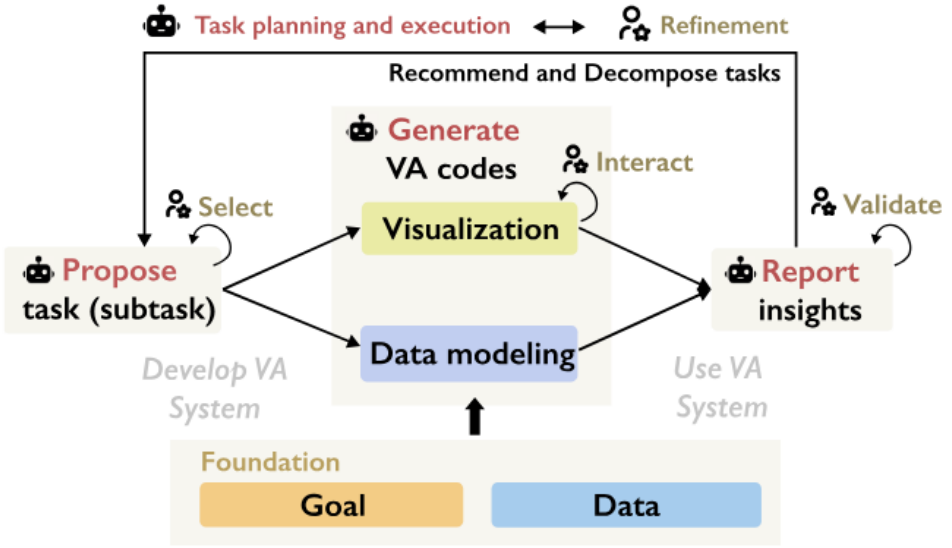
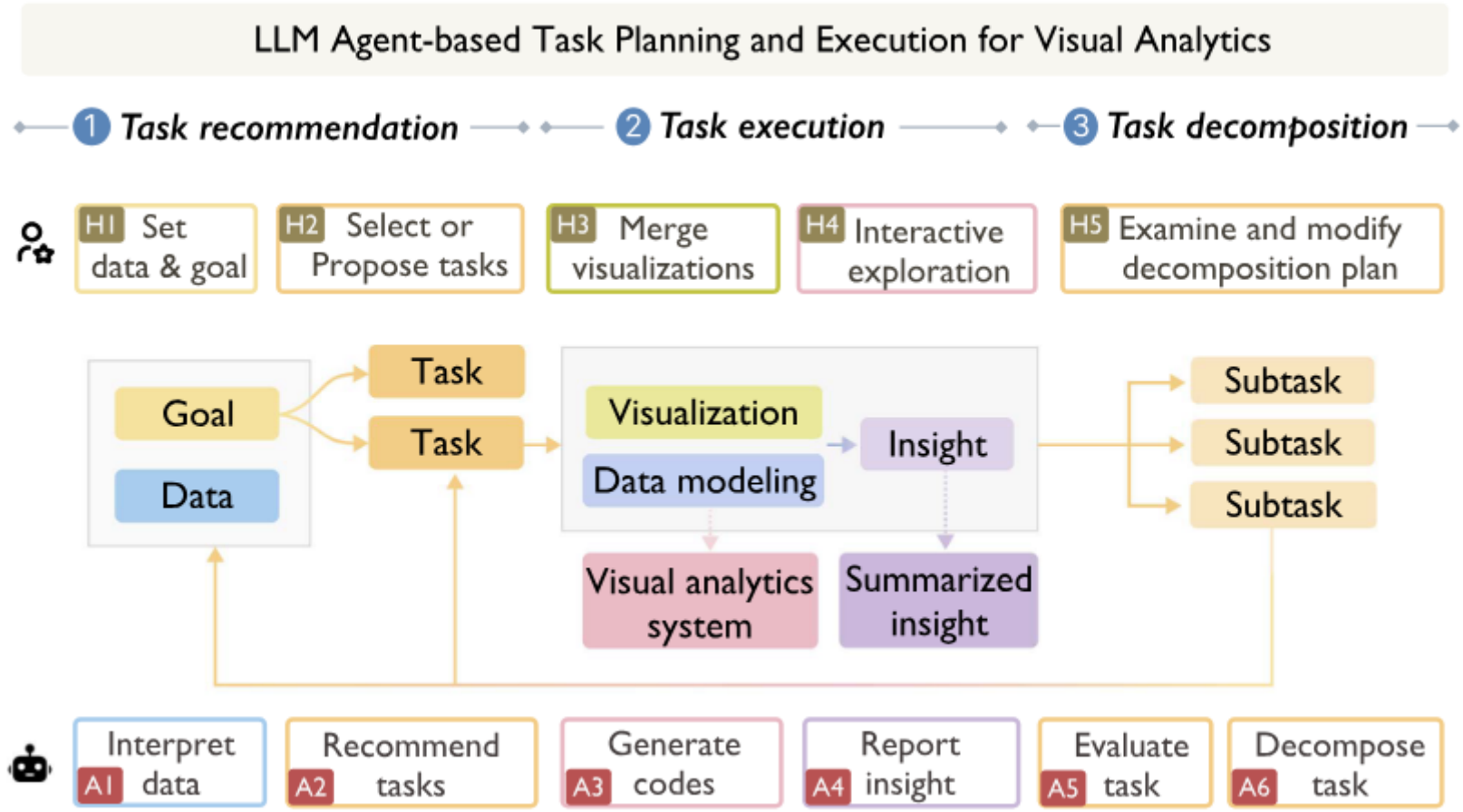
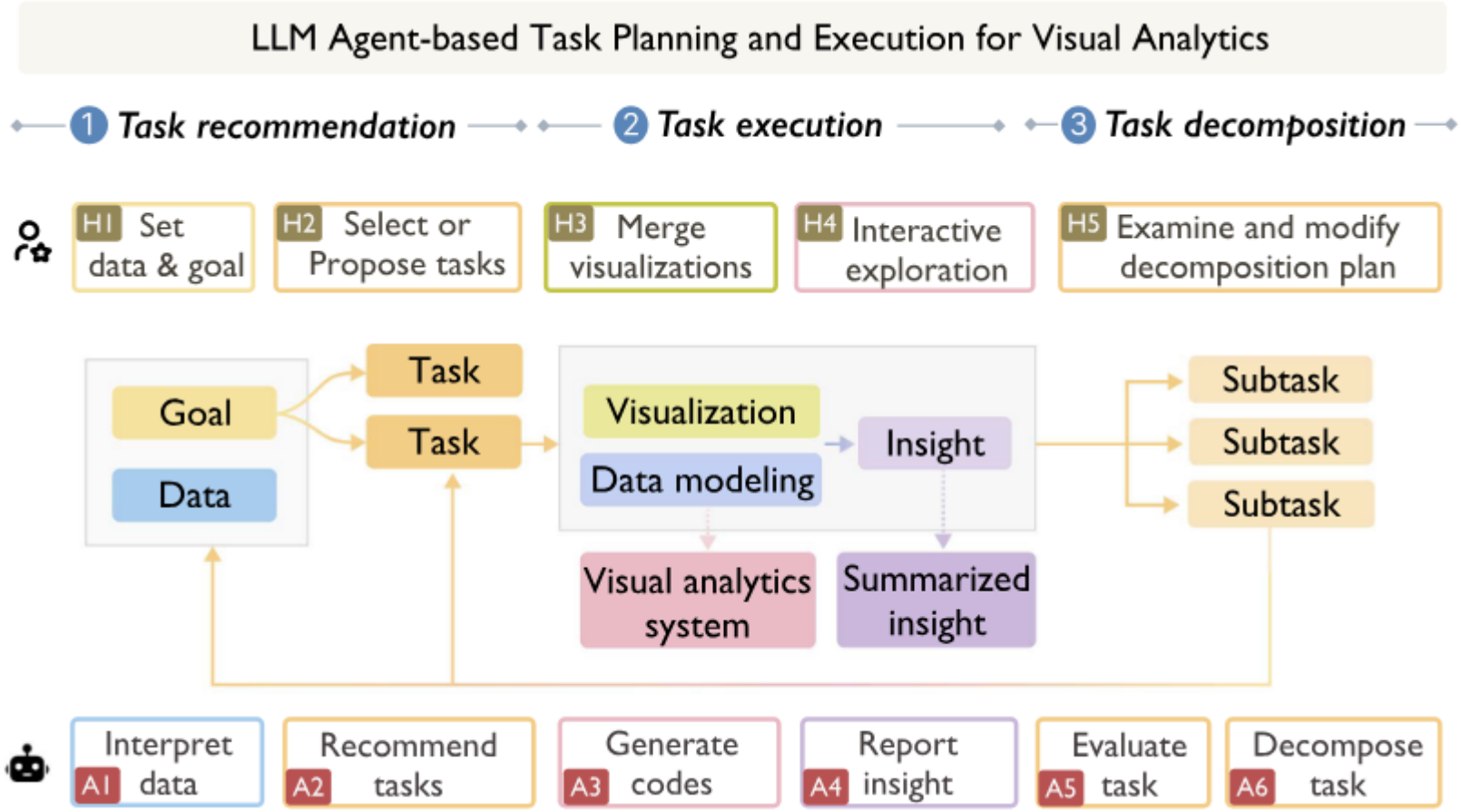


Fig. 1. The illustration of conceptual framework in LightVA. The agent creates the VA system for analysis by task planning and execution, while the user uses the created VA system and refines agent outputs.

LightVA: Collaboration workflow



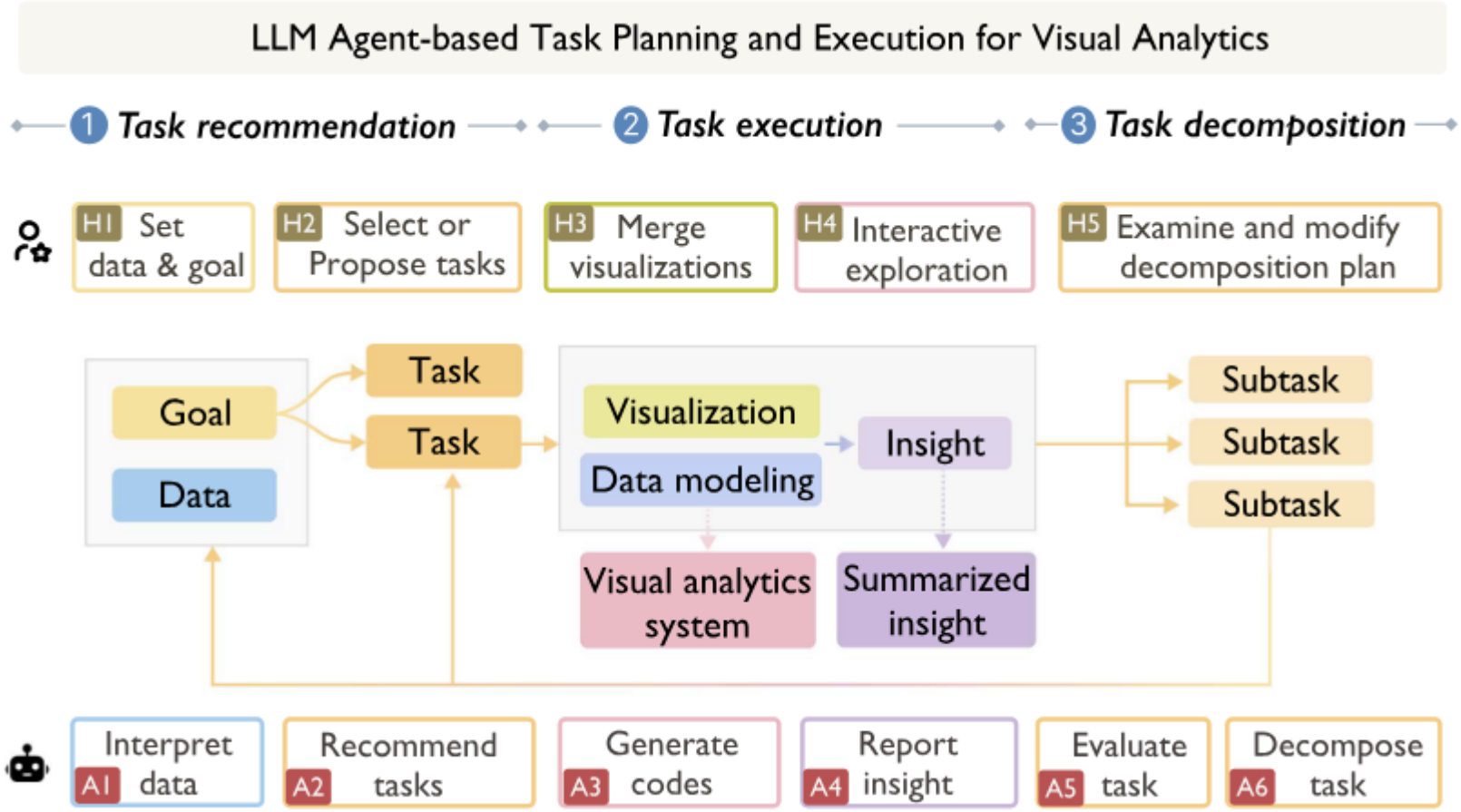
LightVA: Collaboration workflow



Stage 1: Task recommendation

- Human: upload data, state goal, accept / modify tasks
- Agent: interpret data, propose tasks

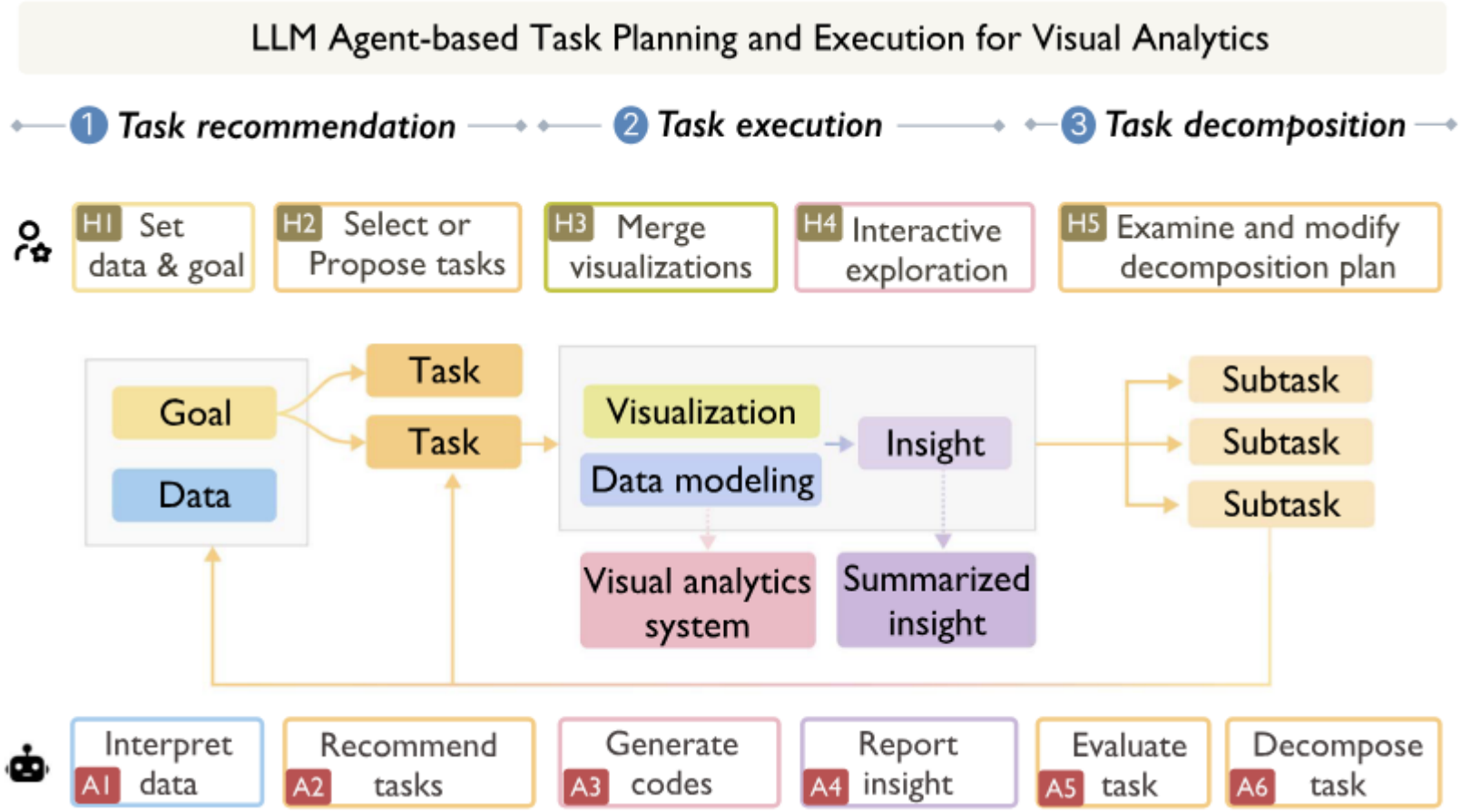
LightVA: Collaboration workflow



Stage 2: Task execution

- Human: merge views, explore linked view
- Agent: generate code for modeling and visualization, summarize insights

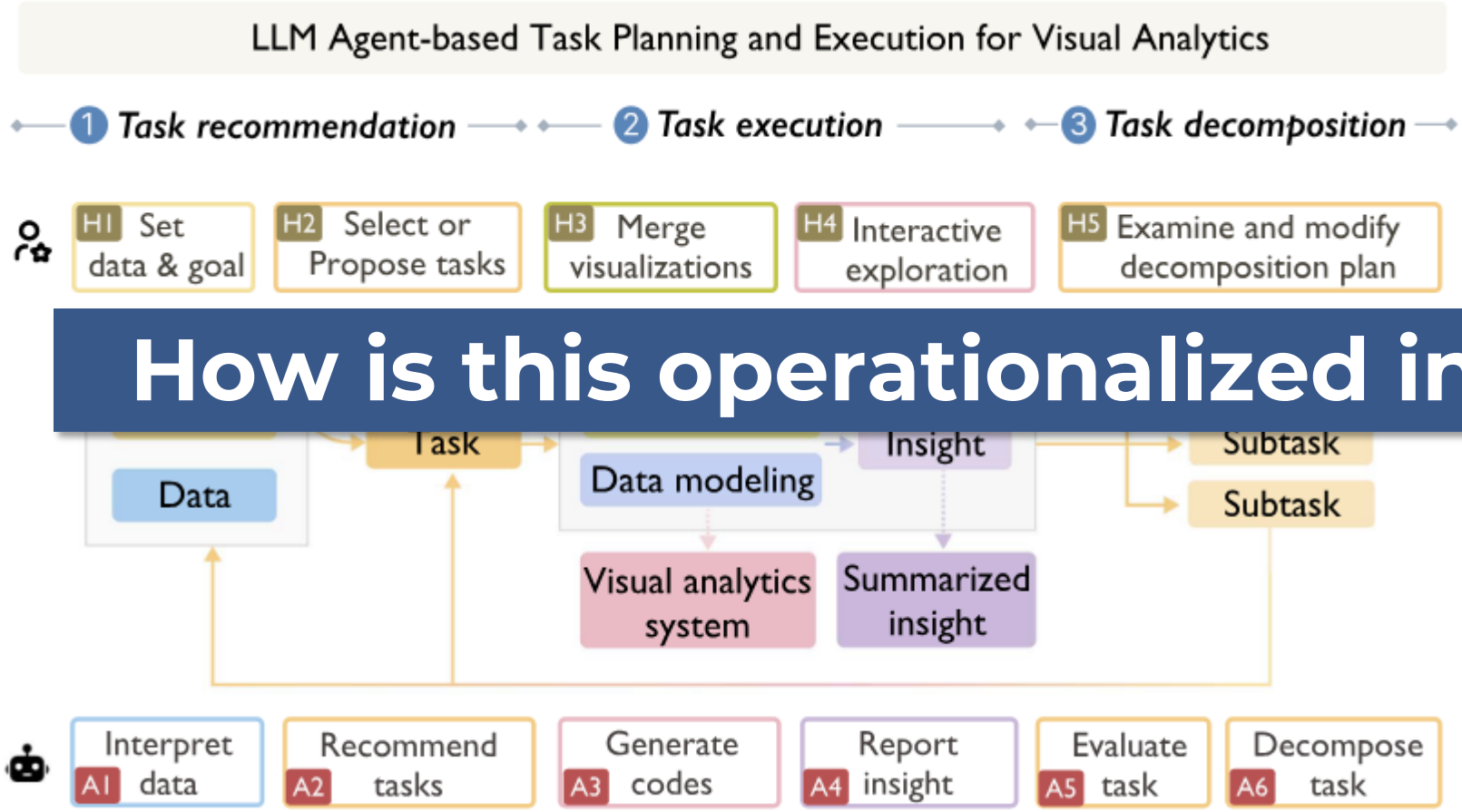
LightVA: Collaboration workflow



Stage 3: Task decomposition

- Human: review plan
- Agent: assess results, propose decomposition plan

LightVA: Collaboration workflow

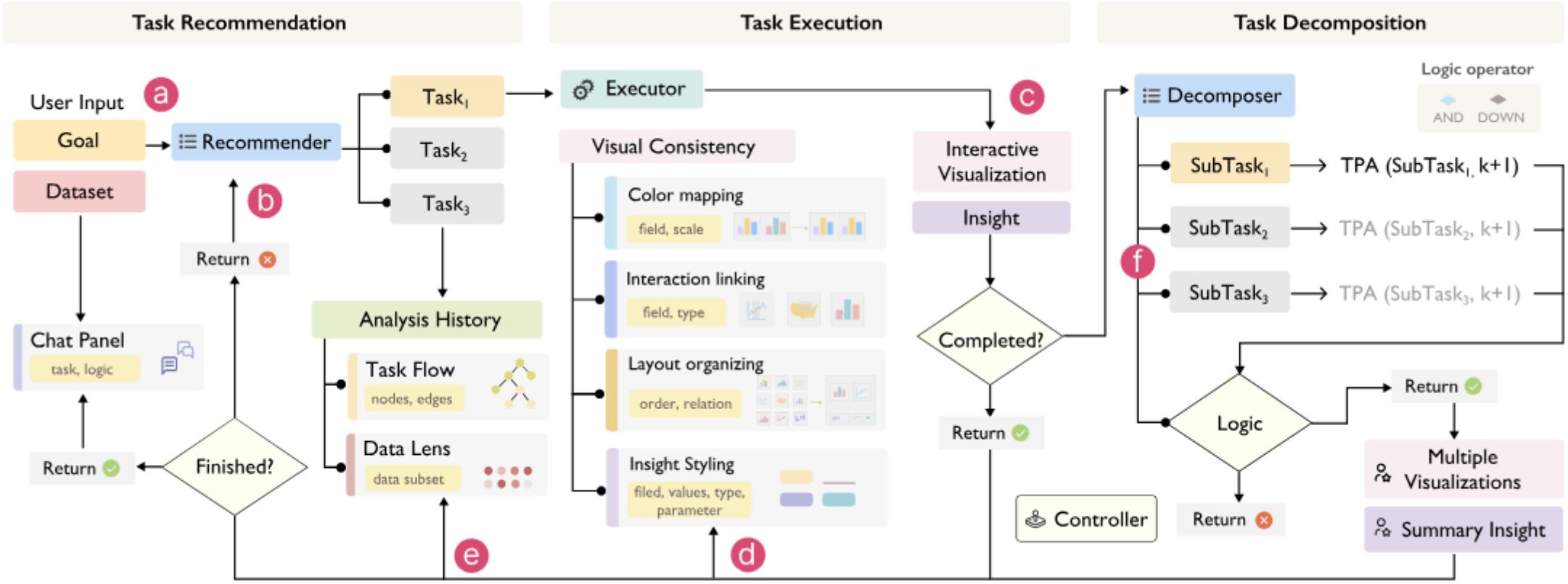


Stage 3: Task decomposition

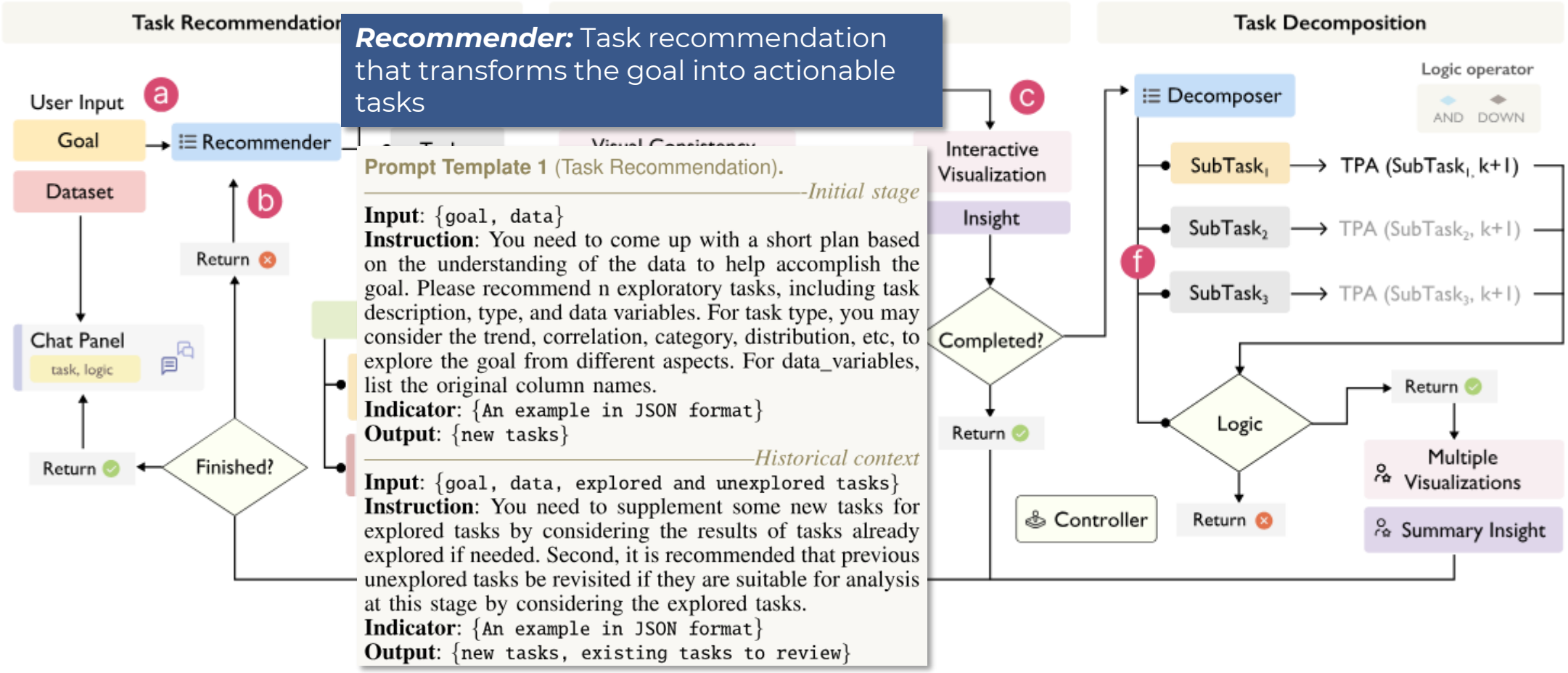
- Human: review plan
- Agent: assess results,

How is this operationalized into a system?

LightVA: System architecture



LightVA: System architecture



Recommender: Task recommendation that transforms the goal into actionable tasks

Prompt Template 1 (Task Recommendation).

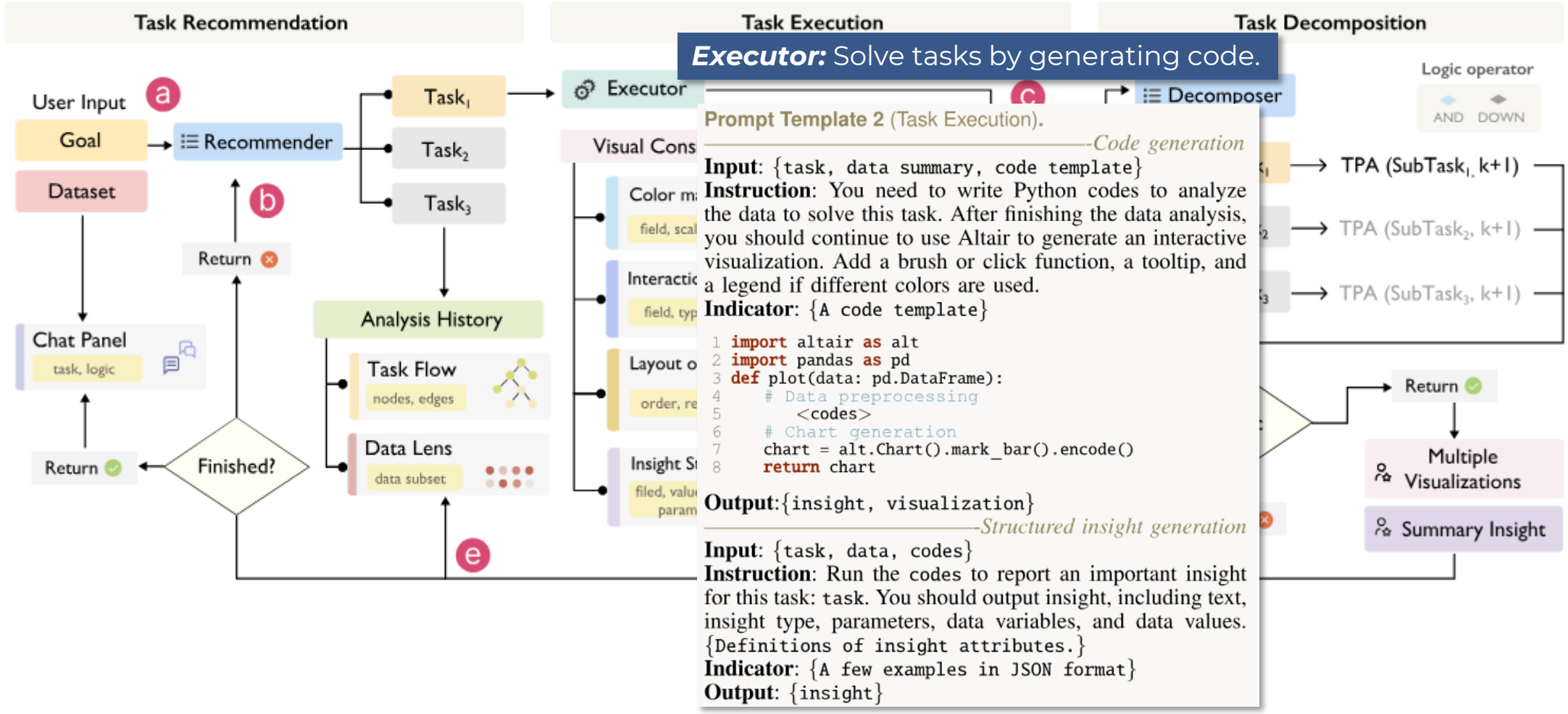
Initial stage

Input: {goal, data}
Instruction: You need to come up with a short plan based on the understanding of the data to help accomplish the goal. Please recommend n exploratory tasks, including task description, type, and data variables. For task type, you may consider the trend, correlation, category, distribution, etc, to explore the goal from different aspects. For data_variables, list the original column names.
Indicator: {An example in JSON format}
Output: {new tasks}

Historical context

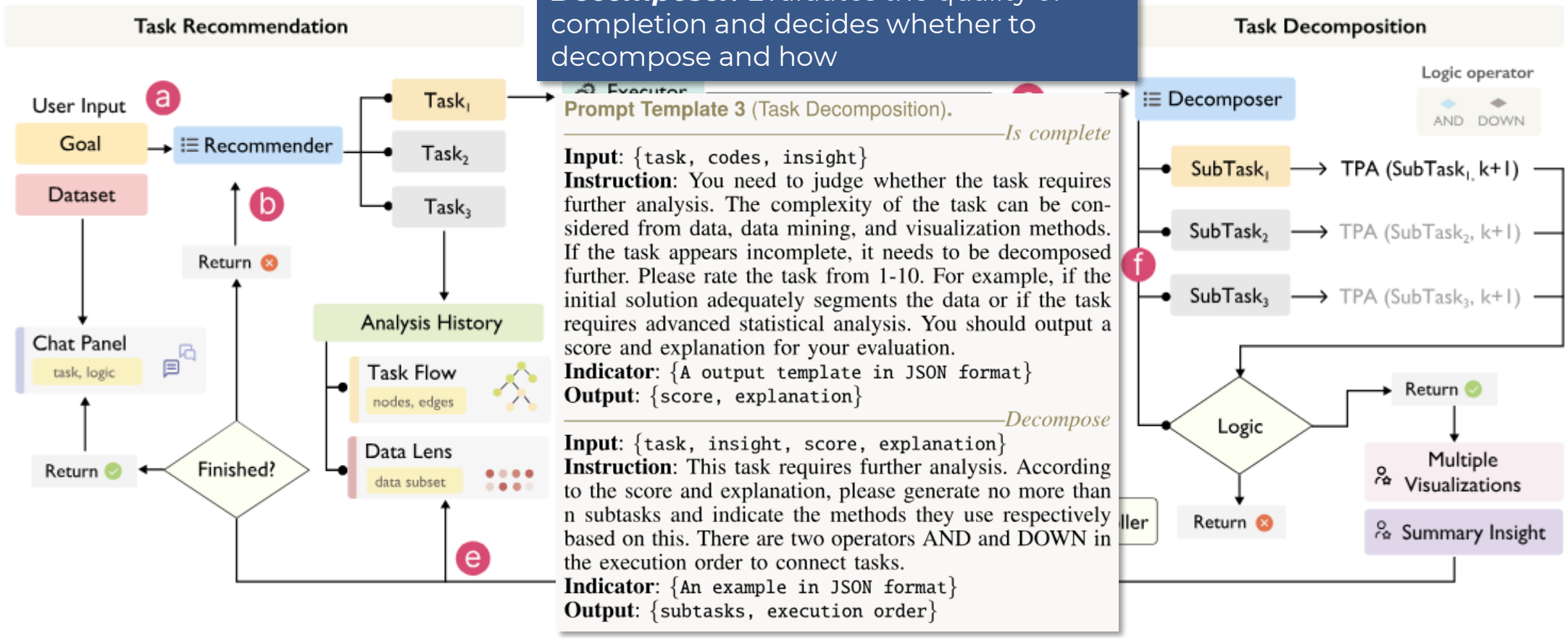
Input: {goal, data, explored and unexplored tasks}
Instruction: You need to supplement some new tasks for explored tasks by considering the results of tasks already explored if needed. Second, it is recommended that previous unexplored tasks be revisited if they are suitable for analysis at this stage by considering the explored tasks.
Indicator: {An example in JSON format}
Output: {new tasks, existing tasks to review}

LightVA: System architecture

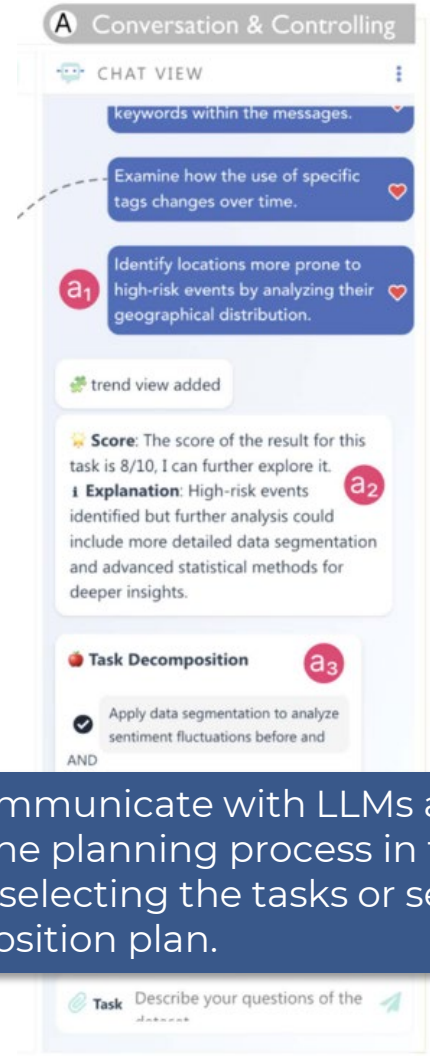


LightVA: System architecture

Decomposer: Evaluates the quality of completion and decides whether to decompose and how

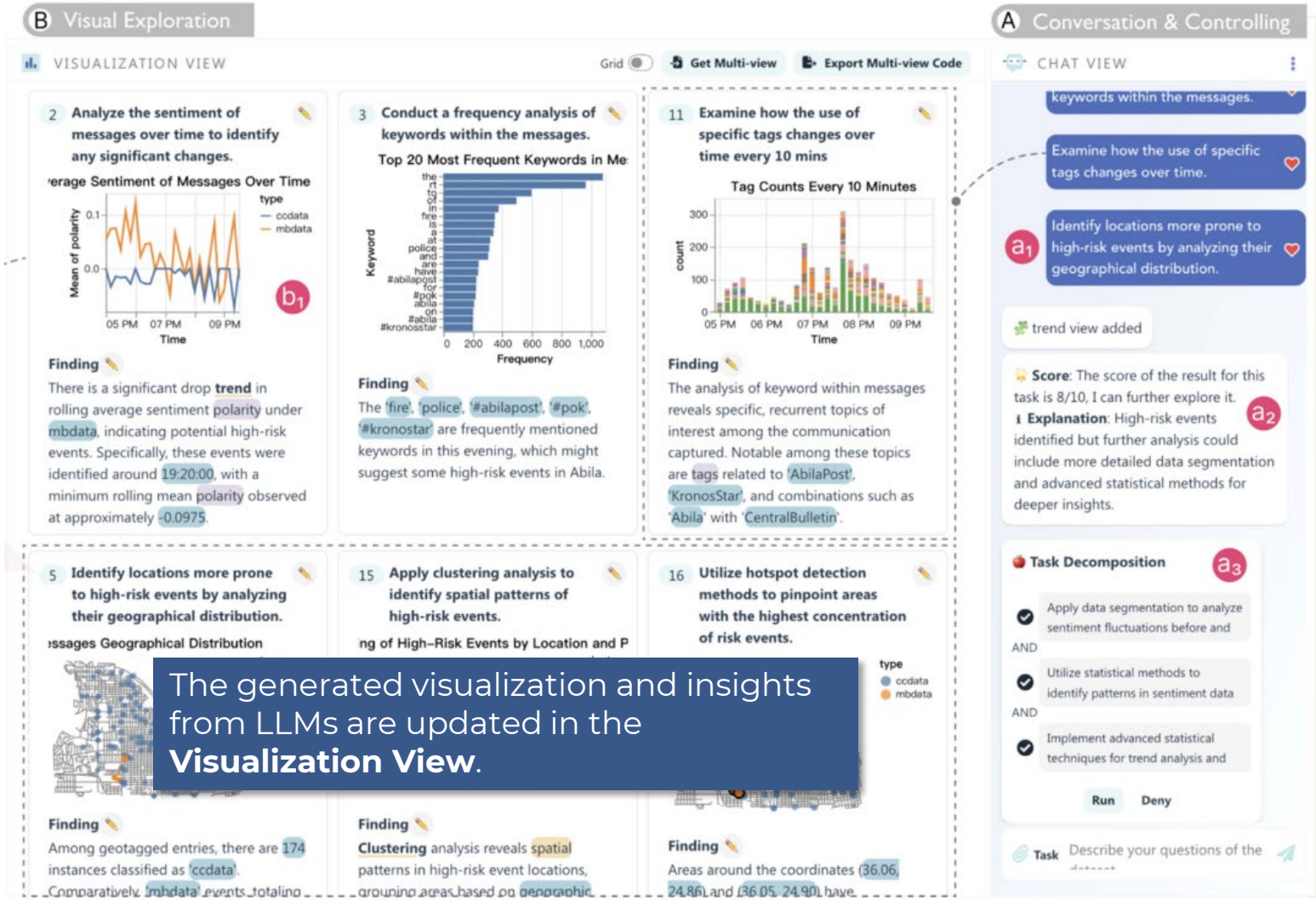


LightVA



Users communicate with LLMs and control the planning process in the **Chat View** by selecting the tasks or setting the decomposition plan.

LightVA



The generated visualization and insights from LLMs are updated in the **Visualization View**.

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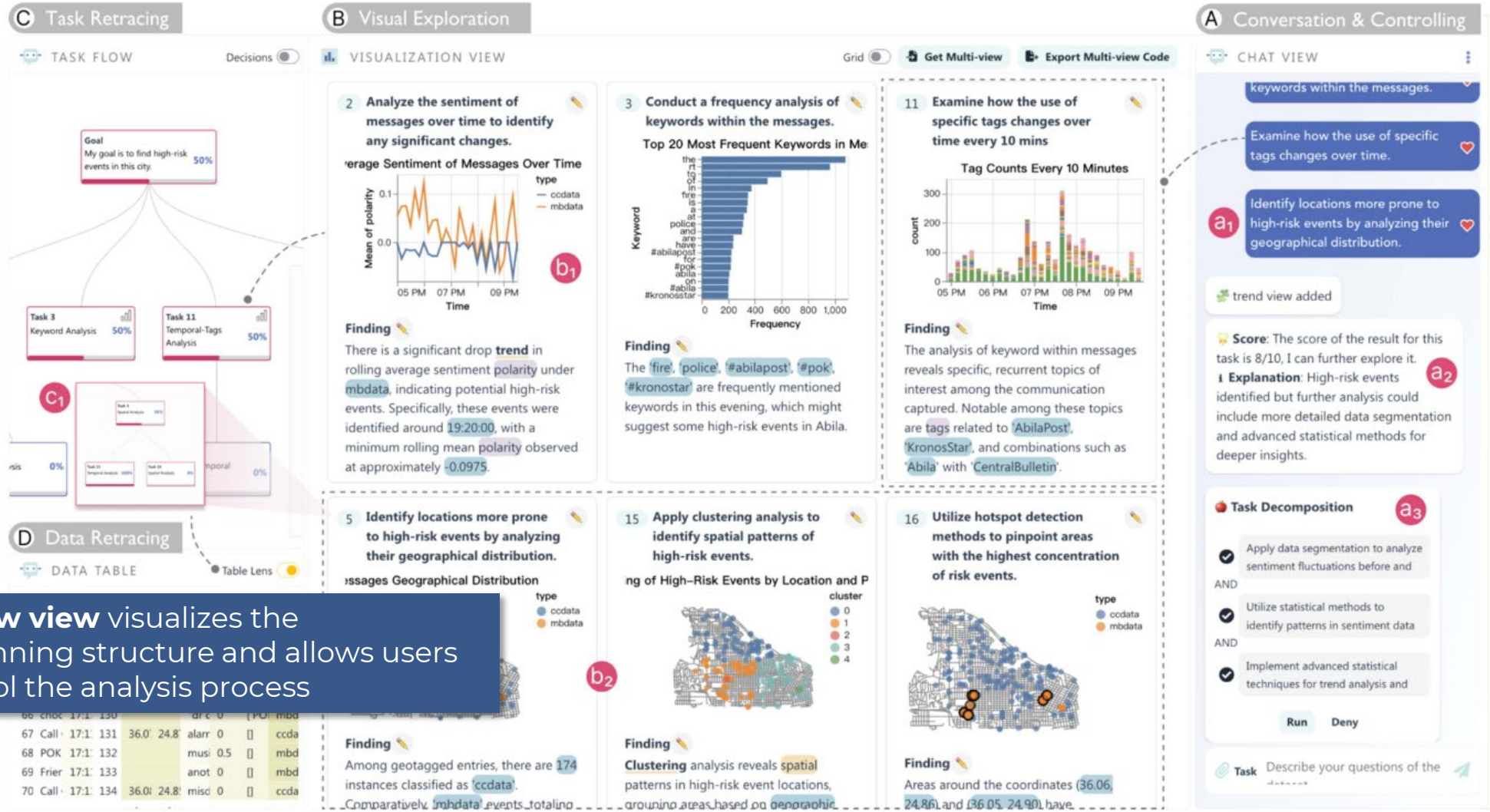
LightVA



Task flow view visualizes the task planning structure and allows users to control the analysis process

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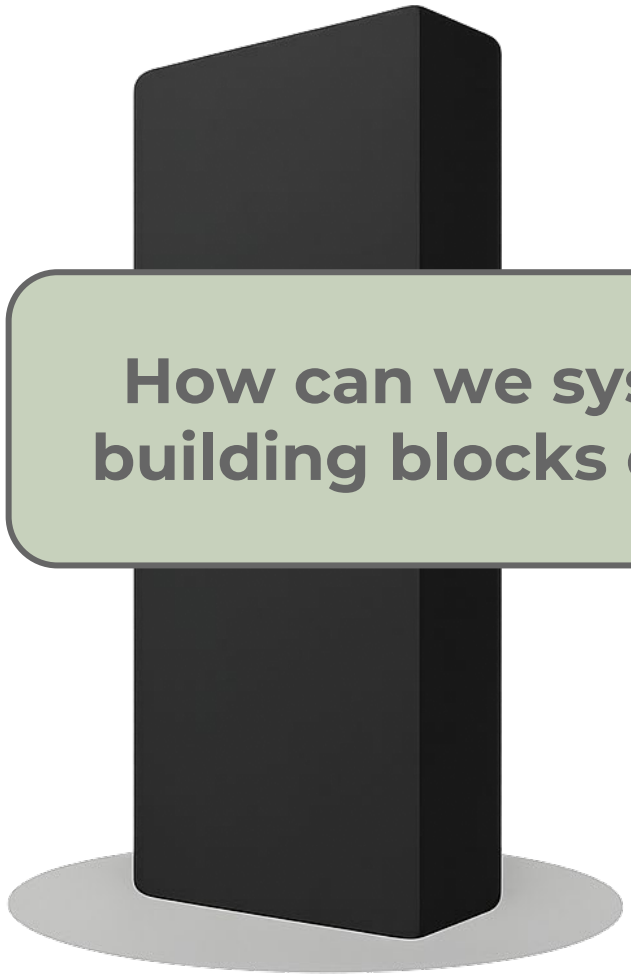
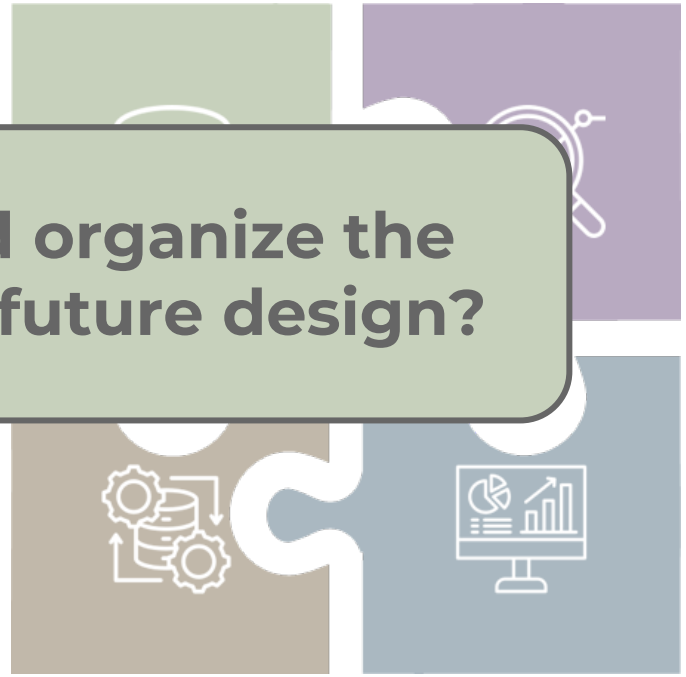
LightVA: Limitations & future work

- **LLM reliability & correctness:** planning, code generation, and insight summaries can be wrong or inconsistent; users need guardrails and verification.
- **Transparency of task planning/decomposition:** users may not understand *why* a task was suggested or decomposed; the rationale can be unclear.
- **Limited "design knowledge" for visualization quality:** chart correctness/clarity/accessibility isn't guaranteed; the system focuses more on "getting something working" than polished design.

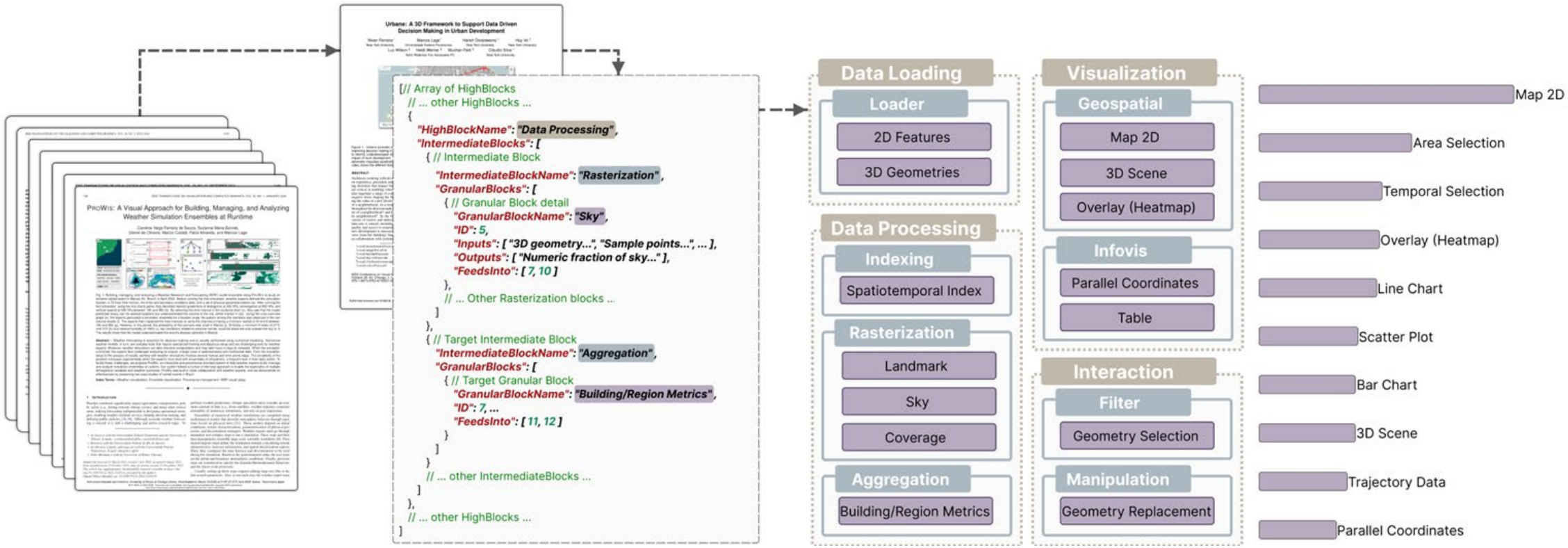
VA-Blueprint



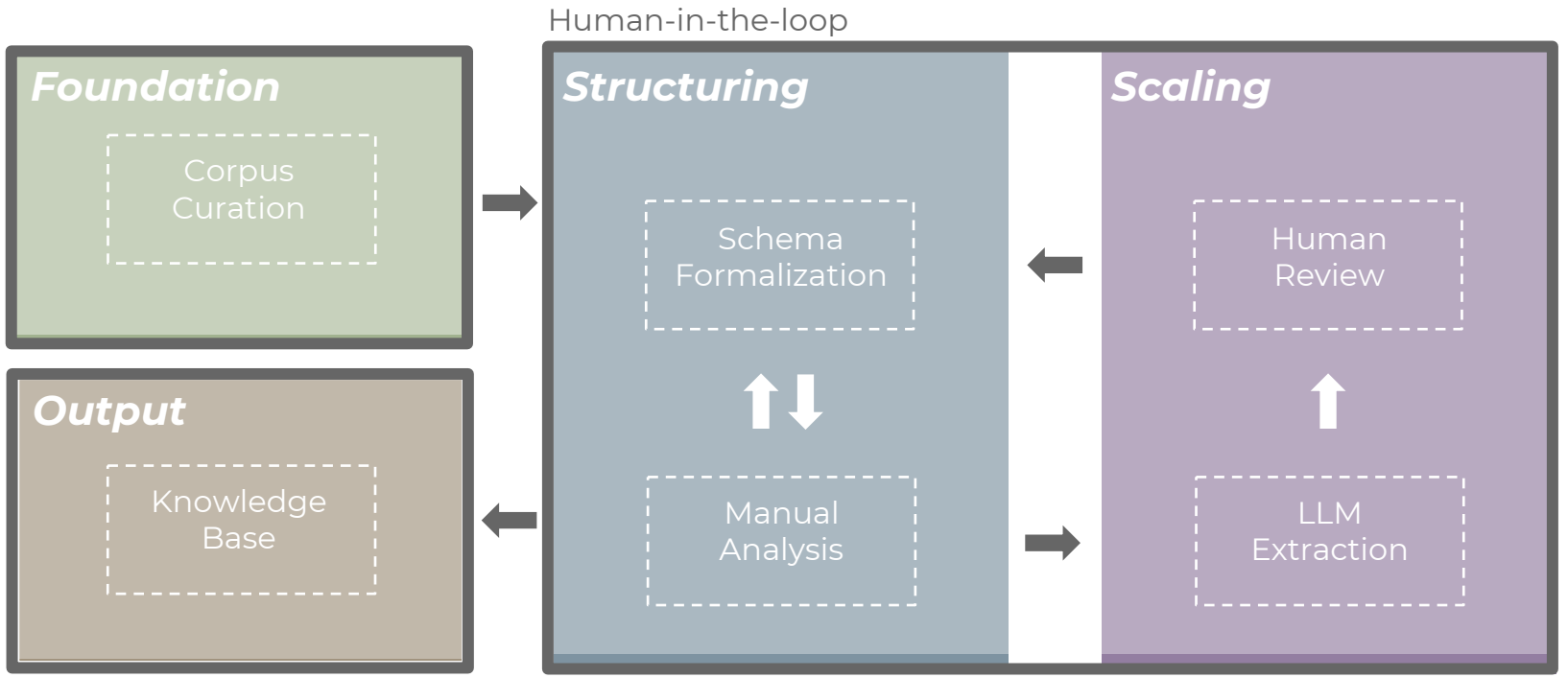
How can we systematically extract and organize the building blocks of VA systems to guide future design?



VA-Blueprint



VA-Blueprint: Overview



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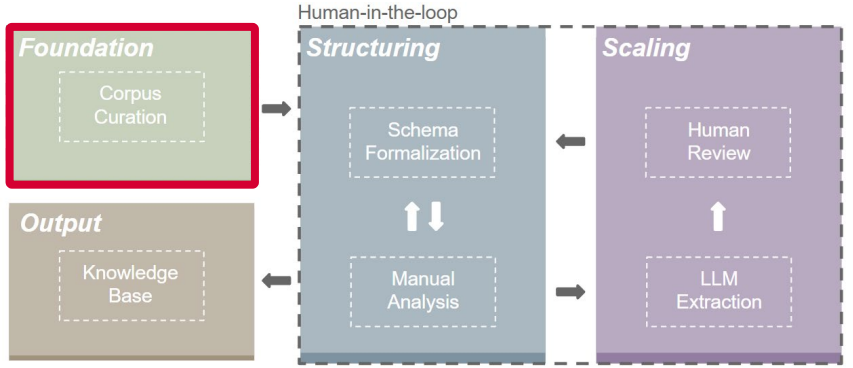

Ferreira et al., 2025

Computers & Graphics 123 (2024) 104013

Contents lists available at ScienceDirect

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journal homepage: www.elsevier.com/locate/cag

Special Section on SIBGRAPI 2023 Tutorials

Assessing the landscape of toolkits, frameworks, and authoring tools for urban visual analytics systems

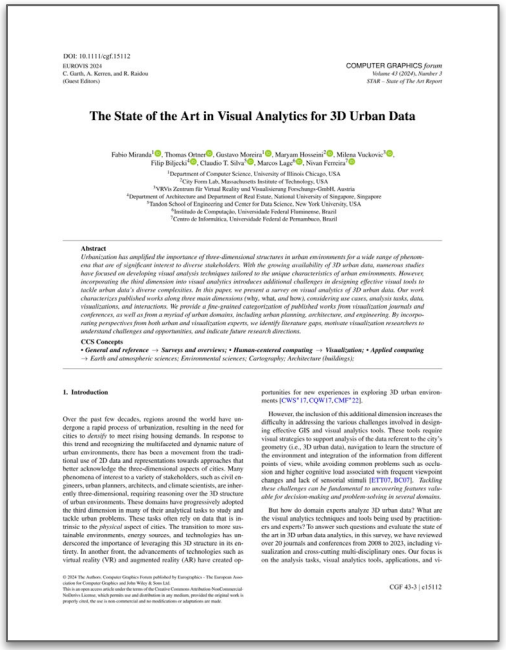
Leonardo Ferreira, Gustavo Moreira, Maryam Hosseini, Marcos Lage, Nivan Ferreira, Fabio Miranda

ARTICLE INFO

ABSTRACT

1. Introduction

Miranda et al., 2024



DOI: 10.1111/cgf.15112

EUROVIS 2024

C. Garth, A. Kerren, and R. Raidou

(Guest Editors)

COMPUTER GRAPHICS forum

Volume 43 (2024), Number 3

STAR – State of The Art Report

The State of the Art in Visual Analytics for 3D Urban Data

Fabio Miranda, Thomas Ortner, Gustavo Moreira, Maryam Hosseini, Milena Vuckovic, Filip Biljecki, Claudio T. Silva, Marcos Lage, Nivan Ferreira

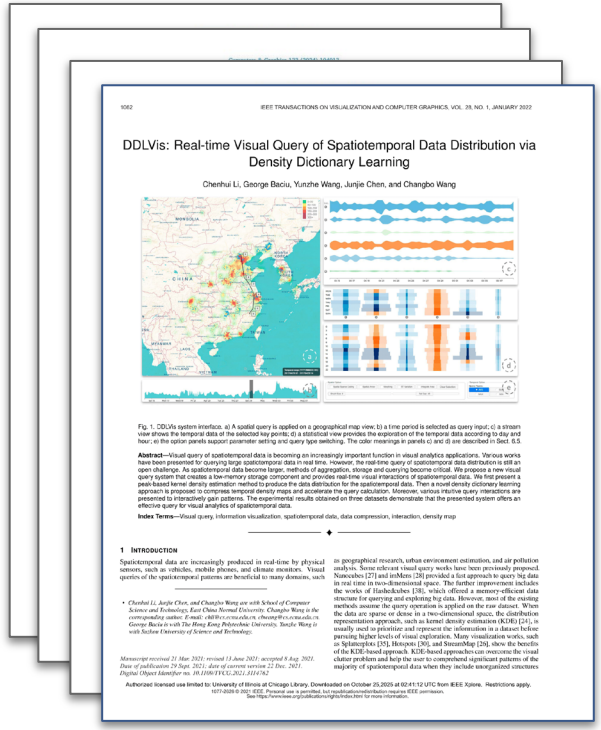
Abstract

CCS Concepts

1. Introduction

CGF 43-3 | e15112

85 papers



1062

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 28, NO. 1, JANUARY 2022

DDLVis: Real-time Visual Query of Spatiotemporal Data Distribution via Density Dictionary Learning

Chenhui Li, George Baciu, Yunzhe Wang, Junjie Chen, and Changbo Wang

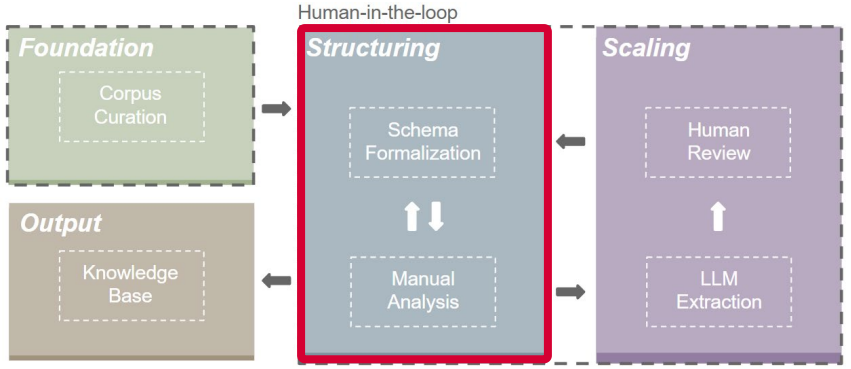
Abstract—Visual query of spatiotemporal data is becoming an increasingly important function in visual analytics applications. Various works have been presented for querying large spatiotemporal data in real time. However, the real-time query of spatiotemporal data distribution is still an open challenge. As spatiotemporal data become larger, methods of aggregation, storage and querying become critical. We propose a new visual query system that creates a low-memory storage component and provides real-time visual interactions of spatiotemporal data. We first present a peak-based kernel density estimation method to produce the data distribution for the spatiotemporal data. Then a novel density dictionary learning approach is proposed to compress temporal density maps and accelerate the query calculation. Moreover, various intuitive query interactions are presented to interactively gain patterns. The experimental results obtained on three datasets demonstrate that the presented system offers an effective query for visual analytics of spatiotemporal data.

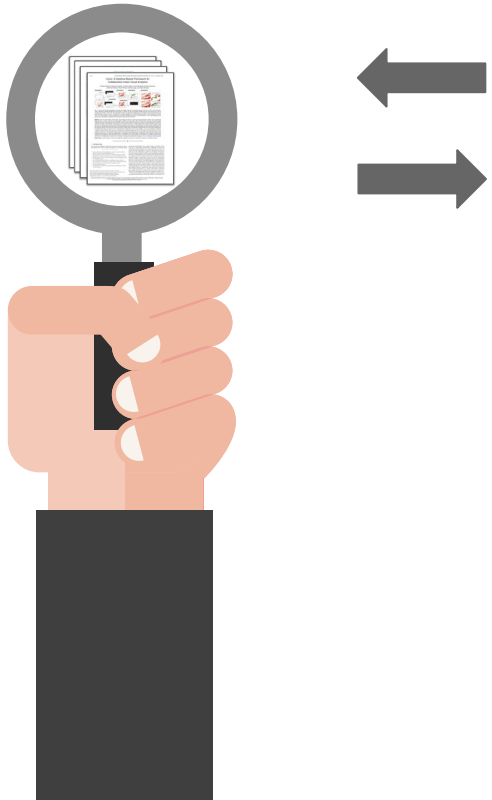
Index Terms—Visual query, information visualization, spatiotemporal data, data compression, interaction, density map

1 Introduction

16 papers

VA-Blueprint: Structuring





VA-Blueprint Schema

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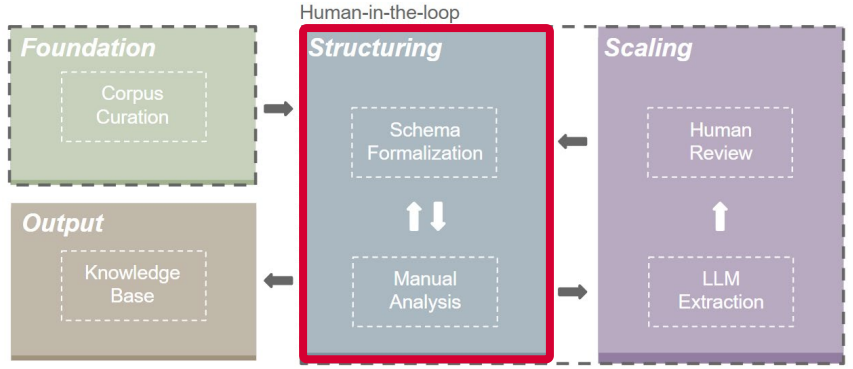
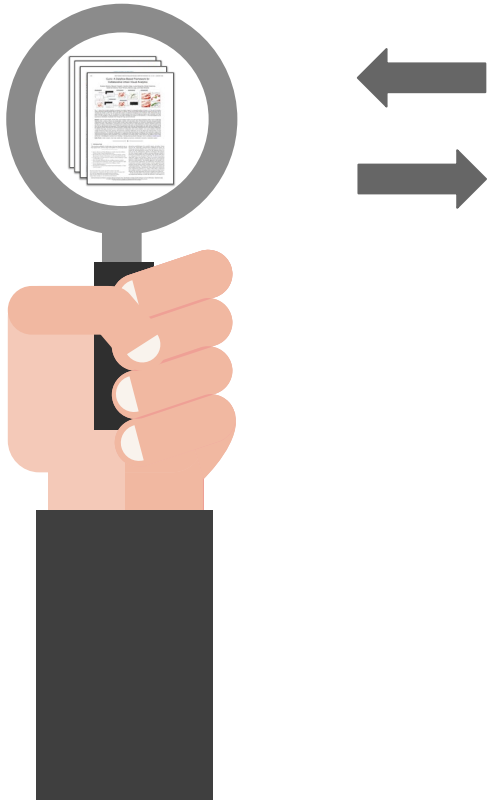
VA-Blueprint: Structuring





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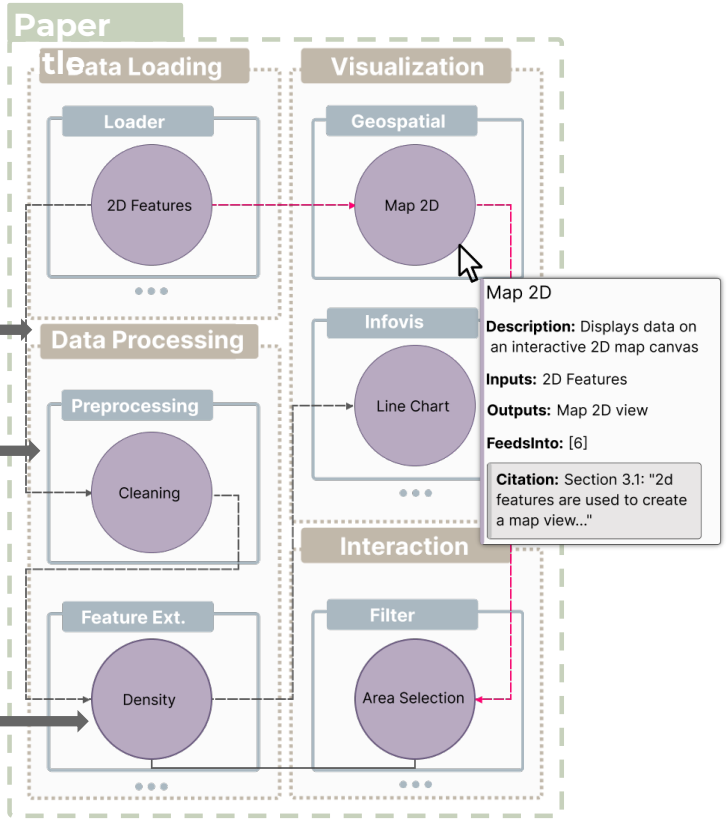

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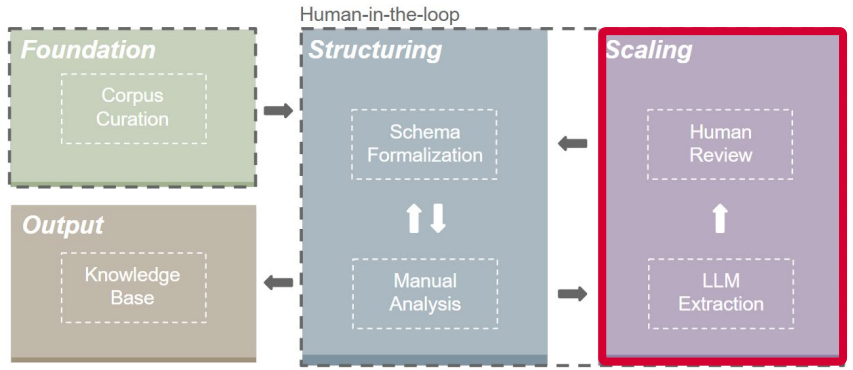

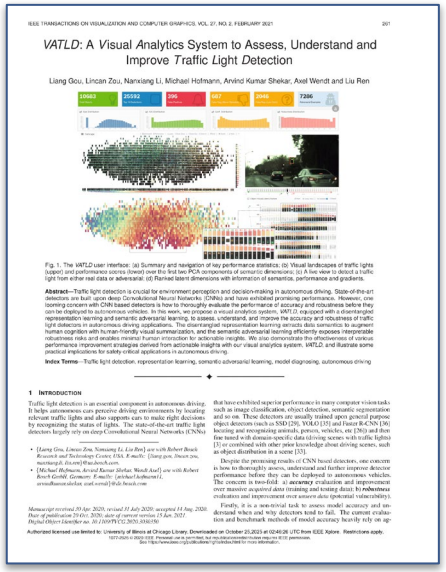

VATLD: A *V*isual *A*nalytics System to Assess, Understand and Improve *T*raffic *L*ight *D*etection

Liang Gou, Lincan Zou, Nanxiang Li, Michael Hofmann, Arvind Kumar Shekar, Axel Wendt and Liu Ren

Fig. 1. The *VATLD* user interface: (a) Summary and navigation of key performance statistics; (b) Visual landscapes of traffic lights (upper) and performance scores (lower) over the first two PCA components of semantic dimensions; (c) A live view to detect a traffic light from either real data or adversarial; (d) Ranked latent dimensions with information of semantics, performance and gradients.

Abstract—Traffic light detection is crucial for environment perception and decision-making in autonomous driving. State-of-the-art detectors are built upon deep Convolutional Neural Networks (CNNs) and have exhibited promising performance. However, one looming concern with CNN based detectors is how to thoroughly evaluate the performance of accuracy and robustness before they can be deployed to autonomous vehicles. In this work, we propose a visual analytics system, *VATLD*, equipped with a disentangled representation learning and semantic adversarial learning, to assess, understand, and improve the accuracy and robustness of traffic light detectors in autonomous driving applications. The disentangled representation learning extracts data semantics to augment human cognition with human-friendly visual summarization, and the semantic adversarial learning efficiently exposes interpretable robustness risks and enables minimal human interaction for actionable insights. We also demonstrate the effectiveness of various performance improvement strategies derived from actionable insights with our visual analytics system, *VATLD*, and illustrate some practical implications for safety-critical applications in autonomous driving.

Index Terms—Traffic light detection, representation learning, semantic adversarial learning, model diagnosing, autonomous driving

1 Introduction

Traffic light detection is an essential component in autonomous driving. It helps autonomous cars perceive driving environments by locating relevant traffic lights and also supports cars to make right decisions by recognizing the states of lights. The state-of-the-art traffic light detectors largely rely on deep Convolutional Neural Networks (CNNs) that have exhibited superior performance in many computer vision tasks such as image classification, object detection, semantic segmentation and so on. These detectors are usually trained upon general purpose object detectors (such as SSD [28], YOLO [35] and Faster R-CNN [36]) locating and recognizing animals, persons, vehicles, etc [26]) and then fine tuned with domain-specific data (driving scenes with traffic lights) [3] or combined with other prior knowledge about driving scenes, such as object distribution in a scene [33].

Despite the promising results of CNN based detectors, one concern is how to thoroughly assess, understand and further improve detector performance before they can be deployed to autonomous vehicles. The concern is two-fold: a) **accuracy** evaluation and improvement over massive *acquired data* (training and testing data); b) **robustness** evaluation and improvement over *unseen data* (potential vulnerability).

Firstly, it is a non-trivial task to assess model accuracy and understand when and why detectors tend to fail. The current evaluation and benchmark methods of model accuracy heavily rely on ag-

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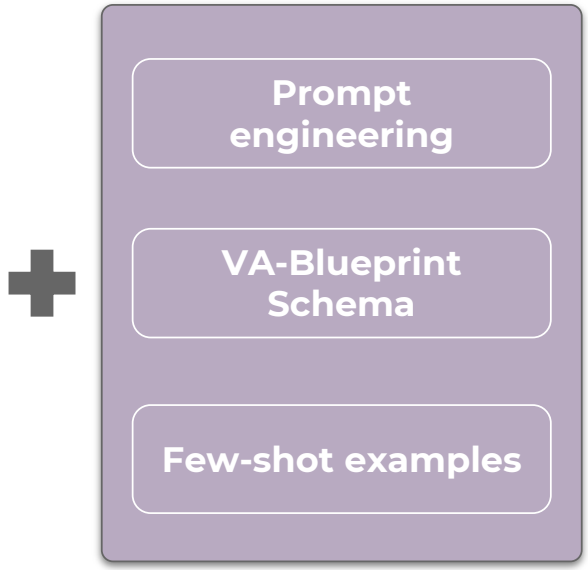
• *{Michael Hofmann, Arvind Kumar Shekar, Wendt Axel} are with Robert Bosch GmbH, Germany. E-mails: {michael.hofmann1, arvindkumar.shekar, axel.wendt}@de.bosch.com.*

Manuscript received 30 Apr. 2020; revised 31 July 2020; accepted 14 Aug. 2020. Date of publication 20 Oct. 2020; date of current version 15 Jan. 2021. Digital Object Identifier no. 10.1109/TVCG.2020.3030470

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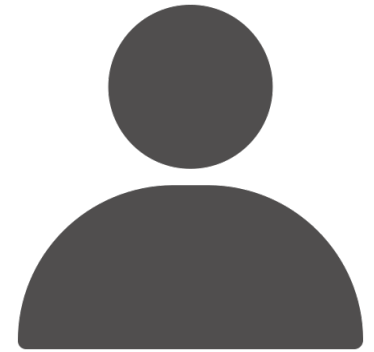
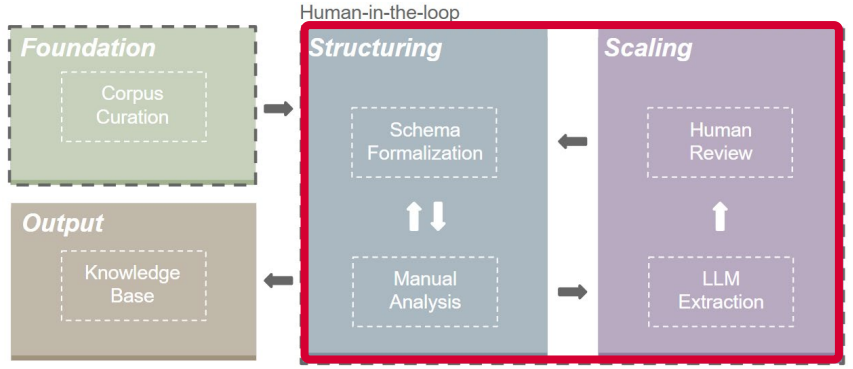
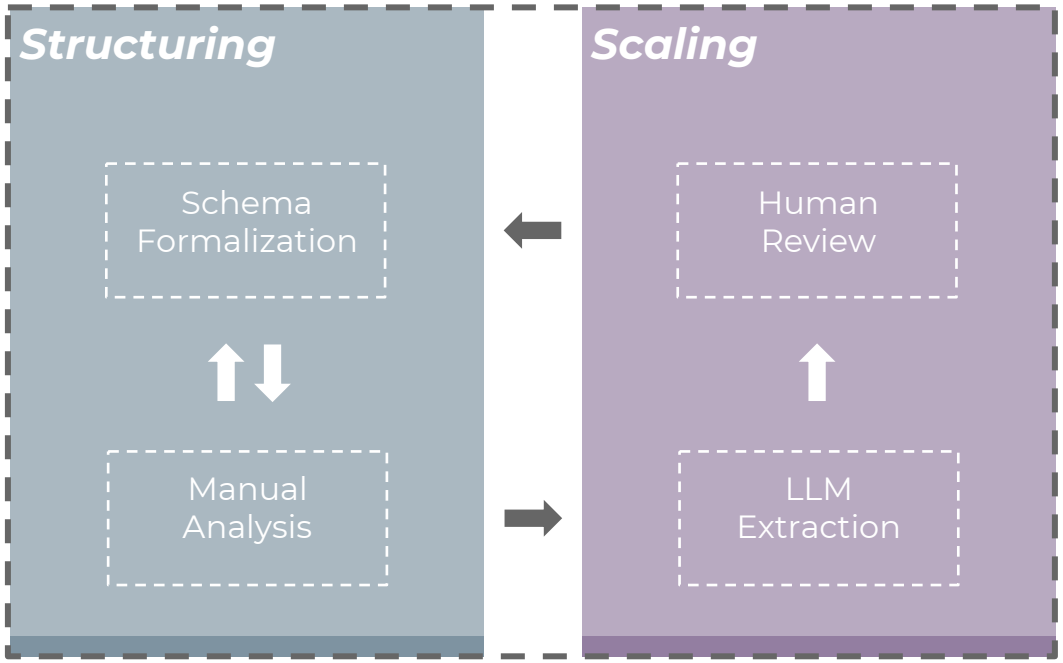
]
```



VA-Blueprint: Human-in-the-loop





VA-Blueprint Schema

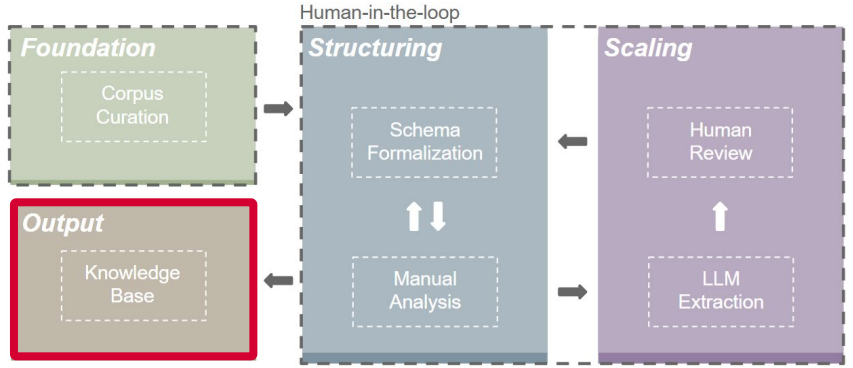
```
SystemBlueprint {
  PaperTitle: string; // Paper Title
  HighLevelBlocks: HighLevelBlock[]; // List of high-level blocks
}

HighLevelBlock {
  HighLevelBlockName: string; // High-level block's name
  IntermediateBlocks: Intermedi                // List of intermediate blocks
}

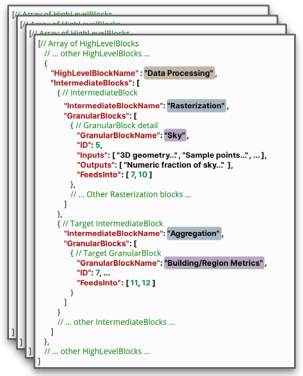
IntermediateBlock {
  IntermediateBlockNam    g; // Intermed    block's name
  GranularBlocks: GranularBlock[]; // List of    ular blocks
}

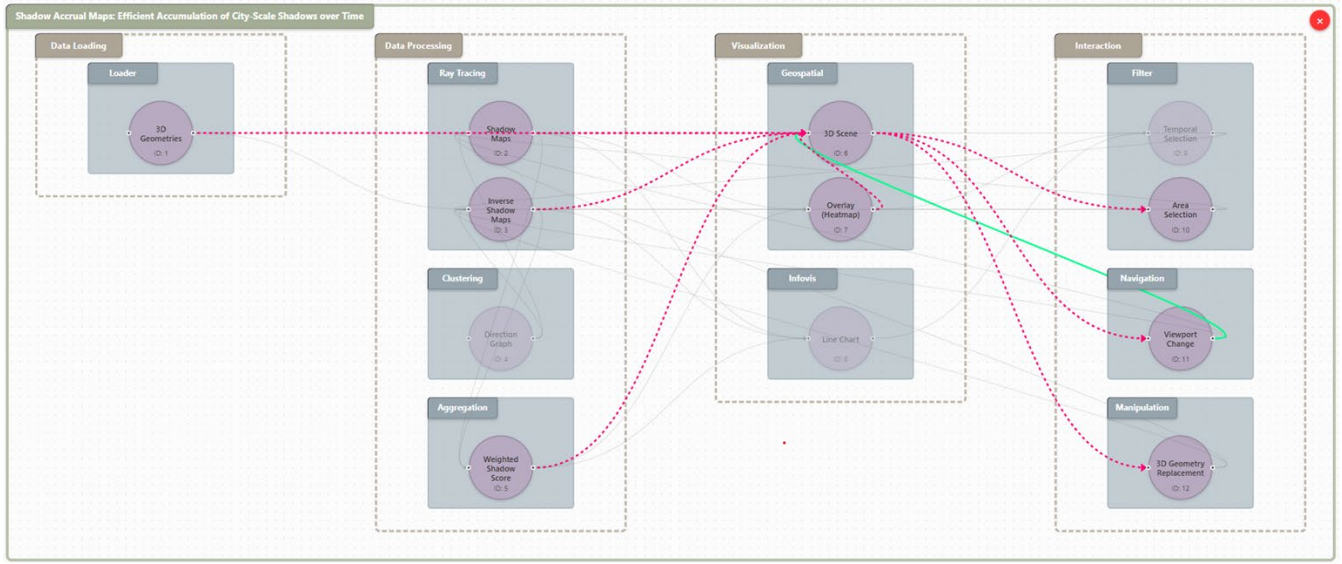
GranularBlock {
  GranularBlockName: string; // Granular block's name
  ID: integer; // Unique block identifier
  PaperDescription: string; // Component's summarized description
  Inputs: string[]; // Data/signals consumed by the block
  Outputs: string[]; // Data/signals produced by the block
  ReferenceCitation: string; // Quote from the paper
  FeedsInto: integer[]; // IDs of downstream blocks
}
```

VA-Blueprint: Output



- Knowledge base with 101 systems
- 2,500+ blocks identified
- 2,400+ connections identified





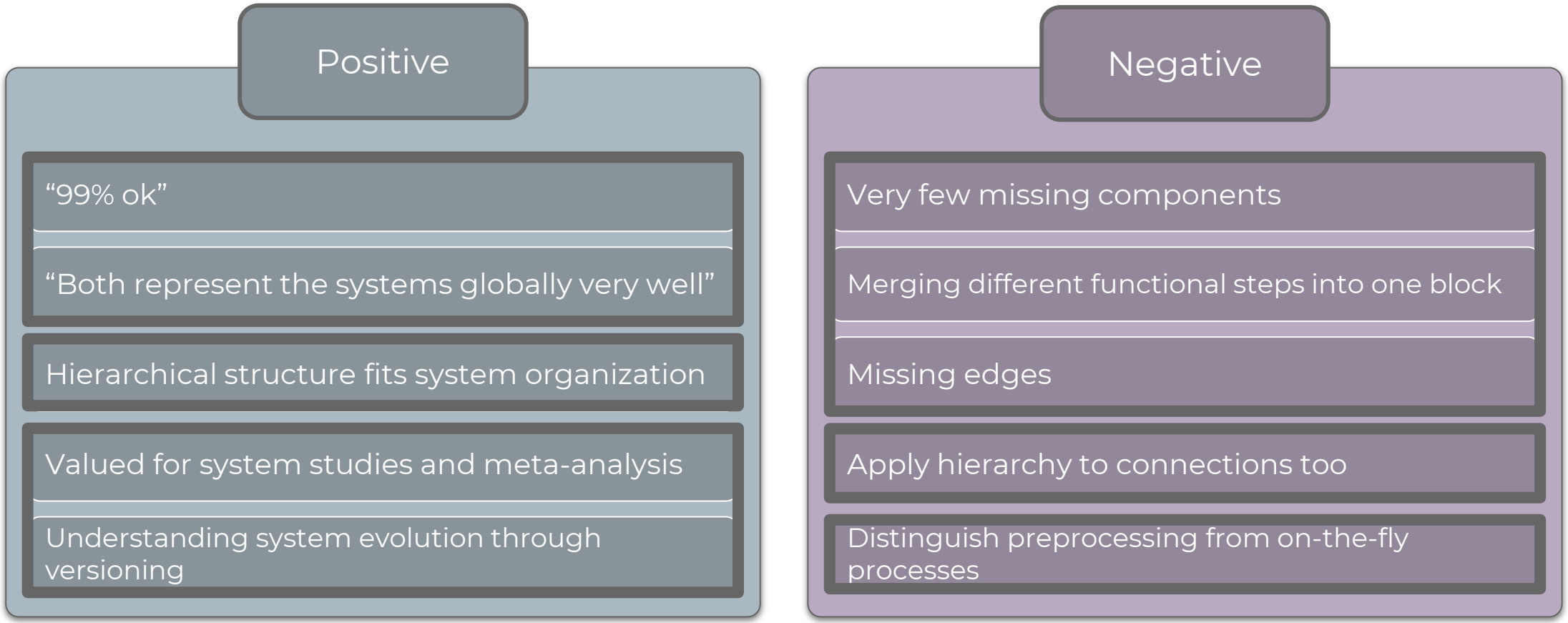
VA-Blueprint: Experts' feedback

accuracy and completeness

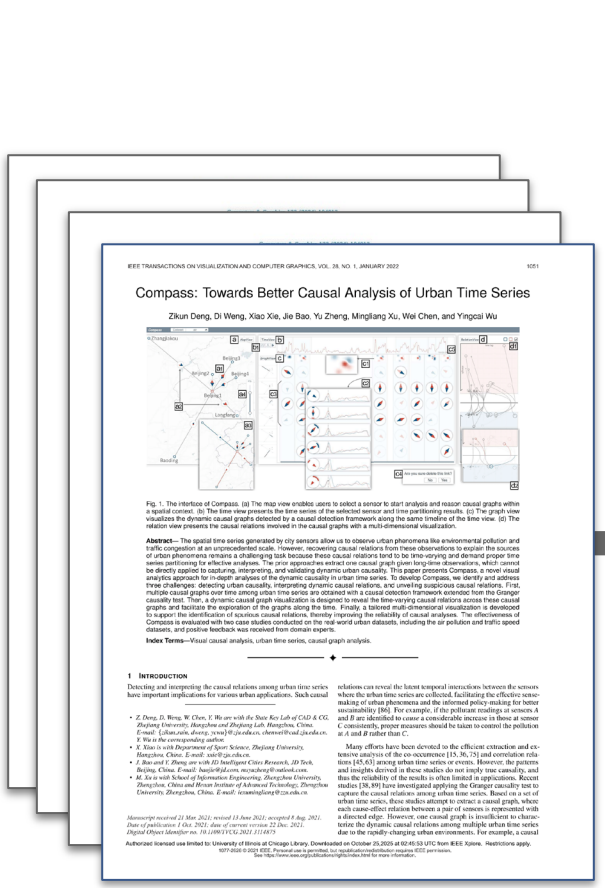
hierarchical representation effectiveness

potential usefulness

VA-Blueprint: Experts' feedback

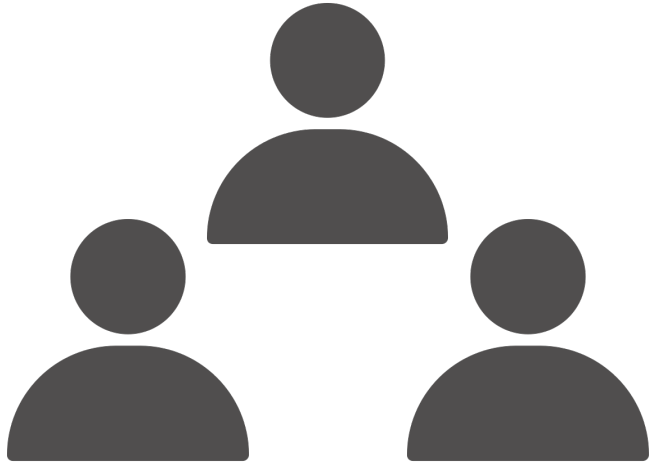


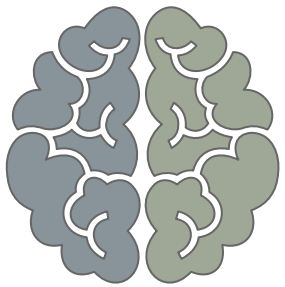
VA-Blueprint: Quantitative analysis



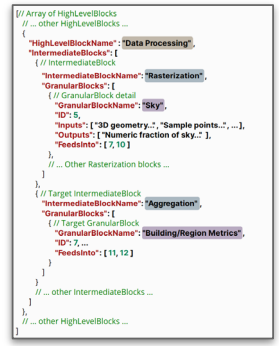
Compass: Towards Better Causal Analysis of Urban Time Series

Zikun Deng, Di Weng, Xiao Xie, Jie Bao, Yu Zheng, Mingliang Xu, Wei Chen, and Yingcai Wu

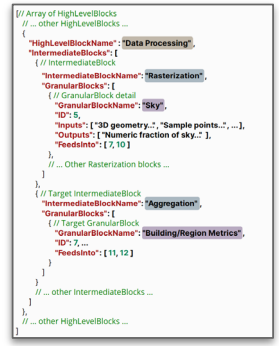




LLM




```
[// Array of HighLevelBlocks
 // ... other HighLevelBlocks ...
 {
  "HighLevelBlockName": "Data Processing",
  "IntermediateBlocks": [
   { // IntermediateBlock
    "IntermediateBlockName": "Rasterization",
    "GranularBlocks": [
     { // GranularBlock detail
      "GranularBlockName": "Sky",
      "ID": 5,
      "Inputs": [ "3D geometry..", "Sample points..", ... ],
      "Outputs": [ "Numeric fraction of sky.." ],
      "FeedsInto": [ 7, 10 ]
     },
     // ... Other Rasterization blocks ...
    ]
   },
   { // Target IntermediateBlock
    "IntermediateBlockName": "Aggregation",
    "GranularBlocks": [
     { // Target GranularBlock
      "GranularBlockName": "Building/Region Metrics",
      "ID": 7, ...
      "FeedsInto": [ 11, 12 ]
     }
    ]
   }
   // ... other IntermediateBlocks ...
  ]
 },
 // ... other HighLevelBlocks ...
]
```


```
[// Array of HighLevelBlocks
 // ... other HighLevelBlocks ...
 {
  "HighLevelBlockName": "Data Processing",
  "IntermediateBlocks": [
   { // IntermediateBlock
    "IntermediateBlockName": "Rasterization",
    "GranularBlocks": [
     { // GranularBlock detail
      "GranularBlockName": "Sky",
      "ID": 5,
      "Inputs": [ "3D geometry..", "Sample points..", ... ],
      "Outputs": [ "Numeric fraction of sky.." ],
      "FeedsInto": [ 7, 10 ]
     },
     // ... Other Rasterization blocks ...
    ]
   },
   { // Target IntermediateBlock
    "IntermediateBlockName": "Aggregation",
    "GranularBlocks": [
     { // Target GranularBlock
      "GranularBlockName": "Building/Region Metrics",
      "ID": 7, ...
      "FeedsInto": [ 11, 12 ]
     }
    ]
   }
   // ... other IntermediateBlocks ...
  ]
 },
 // ... other HighLevelBlocks ...
]
```


VS

Mean

Blocks
Edges

VA-Blueprint: Quantitative analysis

METRIC	MANUAL	LLM
Edges	18.80 ± 6.77	16.65 ± 6.98
Intermediate blocks	7.65 ± 1.76	6.90 ± 1.37
Granular blocks	13.60 ± 3.28	11.30 ± 2.77

Data Processing **22% ⬇**

Of all granular blocks (228), 226 were correct

urbantk.org evl electronic visualization laboratory UIC UNIVERSITY OF ILLINOIS CHICAGO

VA-Blueprint: Limitations & future work

LLM extraction limits	Semantic ambiguity	Domain scope
Issue: Model under-extracts or merges distinct components.	Issue: Some components remain semantically ambiguous or defined at inconsistent levels of detail.	Issue: Current corpus limited to urban VA systems.
Next: Incorporate multimodal extraction, code source, and larger-context models to improve granularity.	Next: Refine taxonomy and hierarchy rules to ensure clearer distinctions and consistent abstraction across components.	Next: Expand to other domains to test schema generalizability.